Automated Sentiment Annotation and Performance Evaluation of Transformer Models on Highly Code-Mixed Malaysian Social Media Text

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Declaration

The project submitted herewith is a result of my own efforts in totality and in every aspect of the project works. All information that has been obtained from other sources had been fully acknowledged. I understand that any plagiarism, cheating or collusion or any sorts constitutes a breach of TAR University rules and regulations and would be subjected to disciplinary actions.



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Abstract

Sentiment analysis tools are typically built and validated on text written in a single standardised language, an assumption that does not hold in Malaysia, where users routinely mix English and Malay within one message. Work here is held back by a shortage of trustworthy labelled data: hand-annotated corpora are reliable but small, while automated pipelines reach social-media scale only by producing a severe neutral-label skew, and existing consensus methods fail because their voters share the same out-of-vocabulary weakness. This project responds with a Heterogeneous Ensemble Annotation Framework. Three models chosen for structural difference, a fine-tuned XLM-RoBERTa encoder with Llama 3.1 8B and Qwen2.5 7B Instruct, read each post and return a label and a confidence score. A confidence-gated routing rule accepts a label only when at least two agreeing voters have each cleared their own confidence threshold; every other post, including a unanimous agreement held weakly, is passed to a regionally adapted adjudicator drawn from a third model family, so that no post is discarded. The framework is applied to a corpus of Malaysian code-mixed posts, with a commitment to annotate at least 150,000 of them. Quality is established in two stages. The complete pipeline is first run over expert-annotated posts held back for the purpose, which is where the confidence thresholds are set and where a weak design would show itself. The finished labels are then compared against a stratified sample of 500 to 1,000 posts relabelled independently and blind by two bilingual annotators, reported as Cohen's Kappa alongside raw agreement and a prevalence-robust coefficient. The corpus is then used to compare a multilingual transformer against English and Indonesian monolingual baselines under Macro-F1, on identical training data and an author-disjoint split, supplying the regional evidence on whether multilingual models hold their assumed advantage on this text type.

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Table of Contents

1 Introduction 2

1.1 Background of the Study 2
1.2 Problem Statement 3
1.3 Research Questions 4
1.4 Proposed Solution and Hypotheses 4
1.5 Research Objectives 5
1.6 Project Scope 6
1.7 Significance and Contribution 7
1.8 Chapter Summary and Evaluation 7

2 Literature Review 10

2.1 Socio-Linguistic Landscape of Malaysian Social Media Discourse 10
2.1.1 Structural Patterns of English–Malay Code-Switching 11
2.1.2 Orthographic Noise and the Limits of Text Normalisation 12
2.2 Corpus Construction and Text Annotation Methodologies (RQ1) 13
2.2.1 Data Annotation Strategies and Trade-offs 13
2.2.2 The Problem of Neutral-Label Skew and False Labels 14
2.2.3 Multi-Model Consensus and Ensemble Voting Logic 15
2.3 Evolution of NLP Architectures for Sentiment Analysis 17
2.3.1 Classical Machine Learning Baselines 17
2.3.2 Encoder-Based Multilingual Transformers 18
2.3.3 Generative Large Language Models (LLMs) 19
2.4 Benchmarking and Evaluation Frameworks in Code-Mixed NLP 21
2.4.1 Evaluation Metrics for Imbalanced Data and Human-Machine Agreement 21
2.4.2 Existing Regional Corpora Benchmarks 22
2.5 Critical Analysis and Summary of Research Gaps 23
2.5.1 Literature Synthesis Matrix 23
2.5.2 Identification of Research Gaps 25
2.6 Summary 26

3 Methodology 28

3.1 The Proposed Heterogeneous Ensemble Annotation Framework 28
3.1.1 Architecture Overview 29
3.1.2 The Three Tiers of Data 32
3.1.3 Design Commitments and Traceability 33
3.2 Corpus and Preprocessing 34
3.2.1 Selection of Target Corpus 34
3.2.2 Corpus Composition and Its Limitations 35
3.2.3 Filtering Design 36
3.2.4 The Normalisation Decision 37
3.3 Heterogeneous Voter Panel 39
3.3.1 Defining Heterogeneity and Its Limits 39
3.3.2 Voter 1: The Fine-tuned Multilingual Encoder 41
3.3.3 Voters 2 and 3: The Generative Voters 43
3.4 Confidence Extraction and Gated Routing 45
3.4.1 Why Agreement Alone Is Insufficient 45
3.4.2 Extracting Confidence from Each Voter 46
3.4.3 The Confidence-Gated Routing Rule 47
3.5 Adjudication 50
3.5.1 Selection of the Adjudicator 51
3.5.2 Adjudication Procedure and Its Limitation 53
3.6 Validation Procedure 55
3.6.1 Stage 1: Pipeline Validation on Held-out Expert Data 55
3.6.2 Stage 2: Human Validation of Framework Output 57

3.6.3 The Dual Role of the Gold Evaluation Set 61
3.6.4 Permitted Use of Each Dataset 62
3.7 Comparative Evaluation (RQ2) 64
3.7.1 Models Compared and the Controlled Variable 64
3.7.2 Split Design and the Author-concentration Constraint 65
3.7.3 Metrics and Reporting Protocol 67
3.8 Experimental Environment and Reproducibility 68
3.8.1 Hardware and Software Environment 68
3.8.2 Running the Annotation 70
3.8.3 What is Pinned and What is Published 71
3.9 Chapter Summary and Evaluation 72

References 77

Chapter 1

Introduction

1 Introduction

This chapter introduces the study. It sets out the background of **sentiment analysis** on Malaysian **code-mixed** social media text, states the problem the project addresses, and presents the research questions, proposed solution, objectives, scope, and significance of the work. The chapter closes with a summary that bridges to the literature review in Chapter 2.

1.1 Background of the Study

Sentiment analysis is the computational task of identifying the emotional polarity of a piece of text, typically as **positive**, **neutral**, or **negative**. It has become a standard tool for analysing public opinion at scale, and it supports applications ranging from brand monitoring to policy evaluation. These applications, however, are typically developed and validated on text written in a single, **standardised language**, and this assumption is precisely what does not hold in the Malaysian context. Malaysia is therefore a demanding case for sentiment analysis, because its online text is routinely **multilingual** within a single message rather than written in one standardised language.

Malaysia is a **multi-ethnic society**, and its online users routinely switch between **English** and **Malay** within a single sentence rather than writing in one language at a time (Treffers-Daller et al., 2022; AlAfnan et al., 2025). The colloquial varieties that result, known as **Manglish** and **Bahasa Rojak**, are not broken English. They are **rule-governed contact varieties** with their own grammar and pragmatic conventions (AlAfnan et al., 2025). This **code-switching** has become a dominant feature of Malaysian social media on platforms such as X, Facebook, and WhatsApp (Treffers-Daller et al., 2022; Maskat et al., 2024), and its use is likely to grow further as the Malaysian digital population is projected to reach around 33.5 million users by late 2025 (Lim et al., 2021).

The scale and informality of this discourse make it a valuable but difficult source of sentiment data. The features that make **Manglish** efficient for people, such as switching language mid-sentence, using sentence-final particles like *lah* and *kan*, and spelling words phonetically, are the same features that disrupt machine **tokenisation** and reduce the accuracy of conventional **Natural Language Processing (NLP)** pipelines built for formal, single-language text (Lim et al., 2021; Mohd Shamsuddin et al., 2024). The consequences are practical as well as academic. Businesses that want to understand customer feedback, and government bodies that want to assess public opinion, are both unable to analyse the sizeable share of Malaysian sentiment expressed in code-mixed form. This project addresses these challenges by investigating **two fundamental issues** in order: first, the **shortage of high-quality labelled data**, and second, the **uncertainty over how different model architectures classify such data**.

1.2 Problem Statement

The difficulty of sentiment analysis on Malaysian code-mixed text comes from **three separate problems** that reinforce one another.

The first problem is that code-mixing and noisy spelling reduce the effectiveness of monolingual NLP tools. Sentiment analysers trained on a single language perform poorly on **Manglish** because they encounter frequent **out-of-vocabulary (OOV)** errors and cannot follow the shifts in meaning that occur across a code-switched phrase (Lim et al., 2021). There are two reasons for this. Structurally, English and Malay are often mixed inside the clause rather than at clean sentence boundaries, so a **tokeniser** has no dependable cue for separating the two languages (Treffers-Daller et al., 2022). In terms of spelling, users write phonetically instead of to a dictionary standard, so the Malay word *tidak* can appear as *tak* or *x*. This inflates the vocabulary and forces **subword tokenisers** to break unfamiliar forms into meaningless fragments (Maskat & Rahman, 2020; Mamta et al., 2023). The affective weight of a message often sits in exactly the tokens these tools discard, such as the Malay particles and slang that appear in no standard lexicon (AlAfnan et al., 2025).

The second problem is that human-validated code-mixed datasets are scarce, and automated attempts to build them introduce a heavy neutral-label skew. High-quality, human-validated sentiment datasets for English–Malay code-mixed text remain critically scarce when compared with monolingual resources (Mohd Shamsuddin et al., 2024). Manual annotation by bilingual experts gives the highest label quality but is too expensive to apply at social-media scale (Kong et al., 2023). Automated efforts to fill the gap tend to produce datasets that are skewed toward the neutral class and that contain "false" labels capable of corrupting model training (Mohd Shamsuddin et al., 2024). The clearest regional example is the **MESocSentiment** corpus, whose semi-automated pipeline produced a distribution of **78.23% neutral, 16.33% positive, and only 5.44% negative**. This skew is a by-product of the labelling method rather than a real property of Malaysian discourse. It is also notable that the skew appeared even though the pipeline already used a **three-model voting ensemble**, because all three voters relied on shallow, non-contextual classification and therefore shared the same **OOV weakness** (Mohd Shamsuddin et al., 2024). Consensus voting among similar models is therefore insufficient; effective correction requires **voters whose strengths complement one another**.

The third problem is that no study has yet tested, in the Malaysian setting, how multilingual transformers compare with monolingual baselines. Transformer-based models have outperformed monolingual baselines on code-mixed tasks elsewhere, but their effectiveness in the regionally diverse Malaysian context has not been examined closely (Mamta et al., 2023). Multilingual models such as **XLM-RoBERTa** are promising for handling linguistic variation,

yet their advantage is not automatic. Conneau et al. (2020) describe a **"curse of multilinguality"** in which a model's finite capacity, once spread across many languages, can allow a monolingual baseline to match or beat a multilingual model on a single-language task. The concern is sharper regionally, since Adilazuarda et al. (2022) found that code-mixing involving Malay was handled noticeably worse than mixing with better-resourced languages. As a result, there is little empirical evidence on how accurately multilingual models classify sentiment in the code-mixed environment of Malaysian social media (Manukonda & Kodali, 2025).

1.3 Research Questions

This project is guided by two research questions.

RQ1: How effectively can a **heterogeneous multi-model consensus annotation framework**, which combines models of different architectures to compensate for one another's linguistic weaknesses, produce a **high-fidelity sentiment-labelled dataset** for code-mixed English–Malay Malaysian social media text, and how closely do its labels agree with human annotation?

RQ2: How do **multilingual transformer-based models** (e.g., **XLM-RoBERTa**) compare with **monolingual baselines** in classifying sentiment in highly code-mixed Malaysian (Manglish) social media text, and how robust are they against the challenges arising from **code-switching** and **noisy spelling variations**?

1.4 Proposed Solution and Hypotheses

To answer these questions, the project proposes a **Heterogeneous Ensemble Annotation Framework** built to generate high-quality sentiment labels for English–Malay code-mixed text.

For **RQ1**, the framework uses a **tri-model majority-consensus mechanism** that deliberately pairs voters of *different* architectures rather than the similar ones used in earlier work. It combines a **fine-tuned multilingual transformer (XLM-RoBERTa)**, chosen for its specialised recognition of regional code-mixed patterns, with two generative Large Language Models from separate developers, Llama 3.1 8B and Qwen2.5 7B Instruct, chosen for their stronger contextual reasoning and zero-shot capability. The transformer is first fine-tuned on existing, unbalanced regional datasets so that it learns Malaysian code-mixed patterns. During annotation, each of the three models reads every post and returns both a predicted label and a confidence score. A label is not accepted on agreement alone. A confidence-gated routing rule accepts it only when at least two agreeing voters have each cleared their own confidence threshold, and every other post, including a unanimous agreement that the voters held weakly, is passed to a separate adjudicator drawn from a third model family and adapted to Southeast Asian languages, so that no post is discarded. This design targets the neutral-label skew directly, because the reasoning of the LLMs

and the pattern recognition of the transformer tend to fail on different examples rather than sharing a single weakness, and because a weakly held agreement is sent for a second opinion rather than certified as consensus.

A **human-in-the-loop (HITL)** validation stage then anchors the framework. **Two bilingual (English–Malay) annotators** independently label a **stratified random sample** of the machine-generated labels (targeted at approximately **500 to 1,000 instances**), and their agreement is quantified using **Cohen's Kappa, with disagreements resolved by discussion to consensus.** This human-validated sample serves as the trustworthy quality benchmark against which the automated labels are compared. In line with this design, the framework is described as producing labels that agree closely with human annotation, with the **human-validated set**, rather than the machine-labelled data, serving as the **gold-standard reference**.

For **RQ2**, the fine-tuned **XLM-RoBERTa** is then evaluated as a sentiment classifier against **monolingual baselines** on the human-validated test set, with a specific focus on **robustness** to code-switching and noisy spelling. This provides the first regional evidence on the multilingual versus monolingual question.

Two hypotheses follow from this design:

H1: A tri-model, **architecturally heterogeneous** majority-consensus framework with human-in-the-loop validation will produce sentiment labels that reach a substantial level of agreement with expert human annotation, higher than the agreement produced by single-model or same-architecture ensemble baselines. (Objectives 2 and 3, RQ1)

H2: A fine-tuned multilingual transformer (**XLM-RoBERTa**) will differ measurably from monolingual baseline models on human-validated Malaysian code-mixed text, measured by **Macro-F1**, with the direction and size of the difference being greatest on instances marked by heavy code-switching and noisy spelling. (Objective 4, RQ2)

1.5 Research Objectives

The aim of this project is to construct a high-quality, human-validated sentiment-labelled Manglish (English–Malay code-mixed) corpus through a heterogeneous multi-model consensus annotation framework, and to determine how multilingual transformer models compare with monolingual baselines for sentiment classification on Malaysian code-mixed text. Four objectives support this aim.

1. To collect and preprocess a corpus of English–Malay code-mixed social media text, including data cleaning, normalisation of noisy spelling where appropriate, and preparation of the raw data for annotation.

2. **(RQ1)** To **design and implement** a heterogeneous tri-model **majority-consensus annotation framework** that combines a fine-tuned multilingual transformer (XLM-RoBERTa), two generative LLMs (Llama 3.1 8B and Qwen2.5 7B Instruct), and human-in-the-loop validation, in order to produce a sentiment-labelled English–Malay code-mixed corpus.

3. **(RQ1)** To measure how closely the framework's automated labels agree with expert human annotation, on a **stratified sample independently labelled by two bilingual annotators**, using inter-rater reliability metrics such as **Cohen's or Fleiss' Kappa**, reported alongside raw percentage agreement and class prevalence, or a prevalence-robust measure such as **Gwet's AC1**.

4. **(RQ2)** To **evaluate and compare** the sentiment-classification performance of a fine-tuned multilingual transformer (**XLM-RoBERTa**) against monolingual baseline model(s) on a human-validated Malaysian code-mixed test set, using **Macro-F1** as the primary metric, and to **assess the robustness** of each model to code-switching and noisy spelling.

1.6 Project Scope

The following boundaries keep the project achievable within the FYP timeframe.

Included in the project. The work targets English–Malay code-mixed text (**Manglish** and **Bahasa Rojak**) drawn from social media. Sentiment is treated as a **three-class polarity task** covering positive, neutral, and negative. The **neutral class** is kept on purpose, so that the neutral-label skew problem can be measured and reduced rather than avoided. The annotation ensemble uses a fine-tuned XLM-RoBERTa together with two generative LLMs, Llama 3.1 8B and Qwen2.5 7B Instruct, deployed in quantised form for computational feasibility, and a SEA-LION adjudicator drawn from a third model family for the posts the ensemble does not settle confidently. Human validation is performed by **two bilingual (English–Malay) annotators** on a stratified random sample of the machine-labelled data, with **inter-annotator agreement** measured by **Cohen's Kappa and disagreements resolved by discussion to consensus before the sample** is treated as the gold standard. For the RQ2 evaluation, **XLM-RoBERTa** is compared against two monolingual baselines: **English BERT (BERT-base)**, representing the English side of the code-mix, and **IndoBERT**, the strongest widely available proxy for the Malay side, given the shortage of dedicated Malay monolingual models and the close relationship between Malay and Indonesian. RQ2 is evaluated on the **human-validated gold test set**, which avoids circular evaluation, since XLM-RoBERTa also serves as an annotation voter in RQ1.

Not included in the project. The work is limited to two languages and does not extend to trilingual English–Malay–Chinese mixing. It does not attempt fine-grained **emotion detection**, **aspect-based sentiment**, or dedicated **sarcasm classification** beyond the three-class scheme. The full training and deployment of downstream production classifiers, together with the complete methodological detail of the framework, are left to the later chapters.

1.7 Significance and Contribution

The project offers value across academic, social, and economic areas, and it aligns with two United Nations Sustainable Development Goals.

Academic contribution (SDG 9: Industry, Innovation, and Infrastructure). In support of **SDG Target 9.5**, which aims to strengthen scientific research and technological capability, the project produces a **gold-standard benchmark corpus** for English–Malay code-mixing. This builds essential **digital infrastructure** for low-resource language research in Southeast Asia. Beyond the dataset, the **architecturally heterogeneous tri-model consensus framework** provides a scalable, reusable method that lets regional researchers move beyond monolingual tools. Researchers in **NLP** and **computational linguistics** are the main beneficiaries, since they gain a validated resource for testing new models and theories of code-switching.

Social and governance contribution (SDG 16: Peace, Justice, and Strong Institutions). In line with **SDG Target 16.7**, which promotes responsive, inclusive, and participatory decision-making, the project addresses a real gap in **inclusive digital governance**. Tools that cannot analyse Bahasa Rojak effectively exclude the voices of citizens who communicate informally. By classifying sentiment in code-mixed discourse more accurately, this work helps bodies such as the **Malaysian Communications and Multimedia Commission (MCMC)** identify public dissatisfaction or social friction, so that national policy can respond to the way people actually write.

Economic contribution. By handling local slang and mixed languages, the framework helps Malaysian **SMEs** and commercial brands interpret customer feedback that standard sentiment tools misclassify. A clearer understanding of consumer grievances supports better service quality and a more responsive digital economy.

1.8 Chapter Summary and Evaluation

This chapter set out the problem that motivates the project. Sentiment analysis on Malaysian code-mixed social media is held back by **three connected issues**: monolingual tools fail against code-switching and noisy spelling, human-validated datasets are scarce while automated alternatives suffer a heavy **neutral-label skew**, and it is still unknown how multilingual transformers compare with monolingual baselines in

this regional setting. In response, the project poses two research questions, proposes a **Heterogeneous Ensemble Annotation Framework** built on a tri-model, architecturally diverse consensus with confidence-gated routing and human-in-the-loop validation, and pursues **four objectives** covering data collection and preprocessing, framework design, agreement measurement, and comparative model evaluation. The next chapter reviews the relevant literature in detail and establishes the foundation on which this solution is built.

Chapter 2

Literature Review

2 Literature Review

Sentiment analysis of **Malaysian social media** sits at the intersection of two hard problems: the **linguistic complexity** of **English–Malay code-switching**, and the scarcity of **high-quality labelled data** needed to model it. This chapter reviews the literature across both, not as a catalogue of individual studies, but as a series of comparisons intended to establish what current approaches can and cannot do, and therefore where this project's contribution lies. **Informal Malaysian discourse** on platforms such as X, Facebook, and WhatsApp is characterised by fluid **multi-ethnic code-switching**, producing the **colloquial varieties** known as **Bahasa Rojak** and **Manglish** (Lim et al., 2021; Maskat et al., 2024). Because conventional **NLP models** are trained predominantly on formal, monolingual corpora, they degrade sharply on this material, suffering **out-of-vocabulary (OOV) errors** and failing to track sentiment across **mid-utterance language boundaries** (Lim et al., 2021; Mohd Shamsuddin et al., 2024).

The review is organised to move from the linguistic problem to the computational response. Section 2.1 characterises the **socio-linguistic landscape**, the **structural patterns of English–Malay switching** and the **orthographic noise** they generate, and argues why these specific features defeat standard **NLP pipelines**. Section 2.2 evaluates **corpus construction** and **annotation methods (RQ1)**, comparing manual, **semi-automated**, and **consensus-based labelling** and the failure modes each introduces. Section 2.3 traces the architectural progression from **classical classifiers** to **multilingual transformers** and **generative LLMs**, weighing the **tradeoffs** of each. Section 2.4 examines benchmarking practice for **imbalanced code-mixed data**, and Section 2.5 synthesises these strands into the specific **research gap** this project addresses.

2.1 Socio-Linguistic Landscape of Malaysian Social Media Discourse

Building a sentiment model for Malaysian social media first requires understanding why the language itself is hostile to standard NLP. Malaysia's multi-ethnic society produces a digital environment in which bilingual users alternate languages within single utterances to optimise efficiency, signal identity, and manage social distance, rather than writing in discrete monolingual streams (Treffers-Daller et al., 2022; AlAfnan et al., 2025). The resulting varieties, Bahasa Rojak and Manglish, are not degraded English but rule-governed contact varieties with their own syntactic and pragmatic conventions (AlAfnan et al., 2025). The sections below establish those conventions, because the features that make this language efficient for humans, mid-clause switching, sentence-final particles, and phonetic spelling are precisely the features that fragment machine tokenisation.

2.1.1 Structural Patterns of English–Malay Code-Switching

Most computational and sociolinguistic work on **intrasentential code-switching** is framed by Muysken's (2013) typology, which distinguishes **insertion**, **alternation**, and **congruent lexicalisation** as distinct **mixing strategies**. The analytically important question for this project, however, is not the taxonomy itself but which patterns dominate **English–Malay text** and why, because different patterns impose different demands on a **tokeniser**. Empirical work by Treffers-Daller et al. (2022) shows that **Malay–English switching** leans heavily toward **insertion** and **congruent lexicalisation** rather than **clause-level alternation**, attributing this to deep **typological compatibility**: both languages share a **Subject–Verb–Object (SVO)** surface order and Malay carries little **inflectional morphology**, so speakers interleave vocabulary **mid-clause** with minimal **grammatical friction**. This matters computationally. Where switching occurs inside the clause rather than at clean clause boundaries, there is no reliable structural cue a tokeniser can use to segment the two languages. This is the first reason monolingual pipelines mishandle the text.

A second, less frequent but more disruptive feature is the heavy use of **closed-class function words** and **discourse particles**. Across most global **code-switching corpora**, **closed-class items** (pronouns, particles, discourse markers) resist switching, but in Malaysian text they do the opposite, serving as habitual **structural anchors** (Treffers-Daller et al., 2022). Particles such as "lah" and "kan" attach to **otherwise-English clauses** ("can or not lah") to soften assertions and signal solidarity, carrying **pragmatic and affective weight** rather than **propositional meaning** (AlAfnan et al., 2025). This is directly consequential for **sentiment analysis**: these particles frequently encode the **emotional stance** of an utterance, yet they appear in neither standard English nor standard Malay lexicons, so **lexicon- and rule-based sentiment tools** discard precisely the tokens that carry tone.

The cost of these patterns becomes visible the moment the text reaches a **monolingual model**. Mohd Shamsuddin et al. (2024) document Malaysian posts whose outer grammar is English while native Malay **noun phrases** are embedded directly into the stream, for example, an English-framed question containing "*institut kanser*" in Malay spelling and word order. A standard **tokeniser** reads such insertions not as **rule-bound bilingual structure** but as broken grammar or **OOV anomalies**, and fragments them into meaningless **sub-units**. In short, the regularities that linguists such as Treffers-Daller et al. (2022) and AlAfnan et al. (2025) describe as systematic are treated by current **NLP tooling** as noise. This gap, between text that is **structurally rule-governed** for humans yet **structurally illegible** to **monolingual models**, is what motivates the **orthographic and architectural problems** examined in the sections that follow.

2.1.2 Orthographic Noise and the Limits of Text Normalisation

Beyond **clause-level structure**, the second barrier to processing **Malaysian social media text** is **orthographic**, and crucially this variation is not random. Maskat and Rahman (2020) propose a formal categorisation of "**Social Media Malay Language**" that resolves the apparent chaos into five recurring patterns: **spelling variations**, **vowel-less words**, **slang-based words**, **loan words**, and **Malay-English mixed sentences**. The practical consequence of this **systematic variation** is a sharp inflation of the **vocabulary space**, in which one **semantic concept** is realised across many **surface forms**, for instance, the Malay negator tidak appearing as tak or x, because users spell **phonetically** rather than to a dictionary standard (Maskat & Rahman, 2020). This **out-of-vocabulary (OOV) inflation**, compounded by emoticons, irregular punctuation, and interleaved **English fragments**, is what renders **noisy Malay text** resistant to **conventional pipelines**, and it remains an actively researched obstacle for **sentiment analysis** (Khalip et al., 2025; Mohd Shamsuddin et al., 2024).

This **surface-form proliferation** defeats both classical and modern **text representations**, but at different points in the pipeline. **Classical feature extractors** such as **Bag-of-Words**, **TF-IDF**, and **static Word2Vec embeddings** treat each **orthographic variant** as an unrelated feature, producing **sparse, fragmented matrices** that sever the link between forms of the same word and fail to capture the underlying sentiment (Selvaraju et al., 2024). **Subword tokenisers** used by **transformer models** do not remove this problem so much as relocate it: confronted with unfamiliar **phonetic spellings**, **Byte-Pair Encoding** and **WordPiece** shatter them into long strings of **atomic sub-units**, inflating **sequence length**, diluting the **self-attention signal**, and causing **semantically identical phrases** to be encoded as unrelated **token sequences** (Mamta et al., 2023; Mohd Shamsuddin et al., 2024). The comparison is the operative point: **classical and neural approaches** fail for the same root reason, dependence on **stable surface forms**, so the **OOV problem** cannot be resolved by model choice alone.

The conventional response is a **normalisation stage** that maps **idiosyncratic variants** back to **canonical forms**, and it is a necessary **preprocessing step** (Mohd Shamsuddin et al., 2024). The literature is candid about its limits, however. Maskat and Rahman (2020) were able to normalise only two of their five categories using **similarity-matching techniques**, reporting that more sophisticated methods are required and that the remaining categories constitute open research. This partial coverage, together with the difficulty **rule- and dictionary-based normalisers** face in keeping pace with continually evolving slang, helps explain the field's recent pivot, exemplified by Khalip et al. (2025), toward models whose **contextual reasoning** can absorb **orthographic variation** directly rather than depending on exhaustive normalisation rules. Preprocessing therefore mitigates but cannot eliminate the **OOV barrier**, which reframes

the central challenge as an **architectural one**: which class of model can natively tolerate this variation? That question motivates the survey of architectures in Section 2.3.

2.2 Corpus Construction and Text Annotation Methodologies (RQ1)

2.2.1 Data Annotation Strategies and Trade-offs

The performance of any sentiment classifier is bounded by the quality and scale of the corpus it learns from, and for low-resource code-mixed languages these two properties pull in opposite directions. Annotation strategies can be arranged along a **fidelity–scalability trade-off** spanning three tiers which are **manual**, **semi-automated**, and **fully automated (generative)** labelling, each buying one virtue at the expense of another.

At one extreme, **manual annotation** by fluent bilingual annotators remains the **fidelity gold standard** but is economically prohibitive at social-media scale. Kong et al.'s (2023) MyCovid-Senti corpus illustrates both sides. Its roughly 11,568 hand-labelled tweets achieve high reliability, yet the labour required to produce them is precisely why manual curation cannot scale to the hundreds of thousands of posts needed for a general-purpose Manglish resource. Manual labelling therefore maximises fidelity while sacrificing **scalability**.

The **semi-automated** tier trades in the opposite direction, using models to accelerate labelling at the cost of introducing **model bias**. Mohd Shamsuddin et al. (2024) exemplify this in constructing MESocSentiment: automated language-identification libraries first filtered code-mixed posts from a pool of 229,566 tweets, after which a **multi-model voting ensemble** assigned sentiment labels (the mechanics of which are examined in Section 2.2.3). This removed the dependence on bilingual annotators and scaled the pipeline dramatically, and such automation is increasingly defensible given evidence that automated annotators can rival human crowd-workers on many labelling tasks (Gilardi et al., 2023). The catch is that the labels are only as competent as the models producing them. This is because this ensemble was assembled from shallow, non-contextual classifiers, it inherited their blind spots and produced a severe **neutral-label skew** (analysed in Section 2.2.2). Semi-automation thus recovers scalability but jeopardises **reliability**.

The **fully automated** tier replaces these shallow models with generative LLMs used directly as annotators, as demonstrated by Negi et al. (2026), who exploit the **zero-shot contextual reasoning** of LLMs to label fine-grained opinion without task-specific training data. This appears to dissolve the trade-off, combining scale with deep semantic understanding, but two limitations complicate that conclusion. First, **single-pass LLM outputs** are prone to

hallucination and stochastic variance, requiring algorithmic adjudication to be trustworthy (Negi et al., 2026). Second, and more decisively for this project, the advantage does not transfer cleanly to low-resource settings; Jadhav et al. (2025) found that even GPT-4o and Llama 3.1 405B underperformed fine-tuned BERT baselines on low-resource annotation by **10.2% and 14.1% accuracy** respectively, concluding that fine-tuned transformers remain essential where linguistic resources are scarce.

Taken together, these three tiers reveal that no single strategy simultaneously secures fidelity, scalability, and contextual robustness for low-resource code-mixed text. Critically, the comparison also shows that **neither fine-tuned transformers nor generative LLMs are individually sufficient**. Transformers offer reliable pattern recognition but lack contextual reasoning, whereas LLMs reason well yet falter on under-resourced varieties. This **complementarity** is exactly what motivates a **hybrid annotation pipeline** pairing the structural reliability of a fine-tuned transformer with the reasoning of an LLM under a consensus mechanism, an argument developed across the sections that follow.

2.2.2 The Problem of Neutral-Label Skew and False Labels

A recurring failure of automated and semi-automated labelling for code-mixed text is **neutral-label skew**, the systematic over-assignment of the neutral class at the expense of genuine positive and negative sentiment. The clearest regional evidence is the MESocSentiment corpus, whose semi-automated pipeline produced a distribution of **78.23% neutral, 16.33% positive, and only 5.44% negative** (Mohd Shamsuddin et al., 2024). A skew this extreme is not a property of Malaysian discourse itself but an **artefact of the labelling method**, and diagnosing its mechanism is a prerequisite for justifying any corrective pipeline.

Two distinct mechanisms drive this skew, and conflating them obscures the problem. The first is a **structural default in lexicon and rule-based tools**. Engines such as VADER and TextBlob derive sentiment from a continuous polarity score and classify any value near zero as neutral (Raees & Fazilat, 2024). When code-mixed input contains **out-of-vocabulary (OOV)** tokens, phonetic Malay spellings, slang, or particles absent from the English lexicon, those tokens contribute no polarity signal, dragging the aggregate score toward zero and producing a **"false neutral"** even for strongly opinionated posts (Mohd Shamsuddin et al., 2024; Selvaraju et al., 2024). The second mechanism is **genuine annotation ambiguity;** across code-mixed corpora, the neutral and "mixed-feelings" categories are consistently the hardest for both human annotators and classifiers, because the boundary between an objective statement and a weakly polarised one is intrinsically blurred (Chakravarthi et al., 2022). Critically evaluating the skew therefore means separating labels that are **truly neutral** from those that are **falsely neutral**, a distinction single-model pipelines are structurally unable to make.

This distinction matters because **false neutrals do not merely inflate one class; they corrupt the others**. Every strongly positive or negative post mislabelled as neutral is a training example stolen from precisely the minority classes a downstream classifier most needs to learn (Mohd Shamsuddin et al., 2024). Because macro-averaged evaluation weights each class equally (Section 2.4), this **minority-class erosion** degrades real performance far more than the headline neutral percentage suggests, while simultaneously biasing the trained model toward predicting neutral at inference.

The conventional remedy, **multi-model consensus**, is necessary but, on its own, insufficient, and the evidence for this is the skew itself. Mohd Shamsuddin et al.'s (2024) pipeline was already a **cross-architecture consensus** of three models (VADER, TextBlob, and pysentimiento), yet it still produced the 78.23% neutral skew. The reason is that voting only cancels error to the extent that the voters fail **independently**; because all three relied on shallow lexical or pattern matching, they shared the same OOV blind spot and agreed on the same false neutrals. Genuine correction thus requires voters with **complementary competencies** rather than merely different implementations, specifically, the regional pattern recognition of a fine-tuned transformer combined with the contextual reasoning of an LLM, whose strengths are complementary rather than redundant (Jadhav et al., 2025). This reframes neutral-label skew not as an argument for "more voting," but as the motivation for the **architecturally diverse** ensemble examined in Section 2.2.3.

2.2.3 Multi-Model Consensus and Ensemble Voting Logic

Sections 2.2.1 to 2.2.2 established that single-model labelling is unreliable and that naive consensus does not automatically repair it. This section evaluates **consensus voting** as a method in its own right, organising the literature along the axis that genuinely separates these frameworks: **how each handles disagreement** between voters. Three approaches, static frequency voting, multi-attempt stabilisation, and generative adjudication, trace an evolution from *discarding* disagreement to *reasoning through* it.

The baseline is **static frequency voting**, exemplified by Mohd Shamsuddin et al. (2024), whose pipeline assigned a label only when a simple majority of three structurally distinct models agreed (Table 2.1).

Table 2.1: Architectural Composition of the Baseline Voting Ensemble (Mohd Shamsuddin et al., 2024)

Voter	**Architecture**	**Sentiment Basis**	**Contextual / Regional Capacity**
VADER	Rule-based	Heuristic polarity rules tuned for general social media	None — no code-mixed awareness
TextBlob	Lexicon-based	Dictionary pattern-matching against a sentiment lexicon	None — English-centric lexicon
pysentimiento	Transformer-based	Pre-trained deep model for social NLP	Limited — not fine-tuned on Malay/code-mixed text

The intent is sound: combining a rule-based, a lexicon-based, and a transformer-based model is meant to ensure the voters do not share systemic errors. In practice, two weaknesses undermine it. First, as Table 2.1 shows, all three voters classify on **surface features** with no regional fine-tuning or deep contextual reasoning, so they in fact **share the OOV blind spot** and converge on the same false neutrals (Mohd Shamsuddin et al., 2024; see Section 2.2.2). Second, frequency voting has **no mechanism for resolving genuine disagreement**: when the three models split one-to-one across positive, negative, and neutral (a **1-1-1 tie**), no majority exists, and Mohd Shamsuddin et al. (2024) were forced to **discard** these posts. This produces **dataset attrition** that is actively harmful, because the discarded ties are disproportionately the ambiguous, heavily code-switched edge cases that a robust classifier most needs to learn. Static voting therefore both retains bad labels and deletes valuable ones.

A first refinement replaces architectural voting with **multi-attempt stabilisation**. Niimi (2024) showed that querying a single locally hosted LLM repeatedly and taking the majority across attempts yields more **reproducible** results than a single inference, by averaging out the stochastic variance of generative output. This is a real reliability gain, but it addresses a different failure: it stabilises *one model's noise* rather than introducing genuine architectural diversity, and offers no principled route through the tie problem.

The most capable approach reframes disagreement as **information to be reasoned over rather than noise to be averaged or discarded**. Negi et al. (2026) deploy an LLM as an **adjudicator** that examines *why* models conflict, interrogating, for instance, whether an apparent

contradiction reflects sarcasm or aggressive Bahasa Rojak, and resolves the case rather than purging it, preserving the ambiguous edge cases that frequency voting loses. Work beyond the regional context corroborates the principle: in a controlled comparison, a disagreement-centric **adjudication** stage allowed even non-reasoning models to match or exceed reasoning models that relied on self-verification alone (Ahtisham et al., 2026). The trade-off is **computational cost**, since adjudication adds inference passes (Ahtisham et al., 2026), and single-LLM adjudication still inherits the low-resource weakness documented earlier (Jadhav et al., 2025).

Arranged along the disagreement axis, these frameworks form a clear progression, **discard → stabilise → adjudicate**, each improving on the last. Yet none combines the three properties the preceding analysis identifies as *jointly* necessary for reliable Malaysian code-mixed labelling: **genuinely complementary voters** (a regionally fine-tuned transformer *and* an LLM, not three shallow models), **reasoning-based adjudication** of disagreement, and a **human-in-the-loop** stage to anchor the most ambiguous cases. Assembling these into a single tri-model, majority-voting pipeline with human validation is the contribution this project advances; the architectural components it depends on are surveyed next.

2.3 Evolution of NLP Architectures for Sentiment Analysis

2.3.1 Classical Machine Learning Baselines

Computational sentiment analysis began with **classical machine learning** classifiers, chiefly **Support Vector Machines (SVM)**, **Naïve Bayes**, and **Random Forests**, paired with **static feature representations** such as Bag-of-Words (BoW) and TF-IDF (Mohd Shamsuddin et al., 2024; Selvaraju et al., 2024). A balanced evaluation must acknowledge their genuine strengths: they are computationally cheap, interpretable, and remain competitive with deep models on formal, monolingual text, where one recent comparison found an SVM trailing a fine-tuned BERT by barely two accuracy points while running orders of magnitude faster (Mulianingrum & Hidayat, 2025). For code-mixed Malaysian text, however, two **architectural deficits**, not merely the orthographic noise discussed in Section 2.1.2, render them structurally inadequate.

The first deficit is the **absence of contextual representation**. BoW and TF-IDF reduce a post to token-frequency counts, discarding word order entirely, while even Word2Vec embeddings assign each word a **single fixed vector regardless of context** (Selvaraju et al., 2024). Yet sentiment is **compositional**, it depends on negation, intensification, and polarity accumulated across a clause, precisely the signals these representations erase. This is especially damaging in Bahasa Rojak, where affective weight frequently sits in the sentence-final particles and mid-clause switches identified in Section 2.1.1, which frequency-based features cannot capture. The second deficit is the **lack of cross-lingual transfer**: a classical model has no mechanism to

relate tokens from one language to the other, so each language fragment is learned in isolation rather than interpreted as part of a single bilingual utterance.

Crucially, the impact of these deficits is **measurable rather than asserted**, and the evidence speaks directly to RQ2. On Hindi–English code-mixed data, Wadhawan and Aggarwal (2021) report that an SVM baseline reached only **58.2% accuracy** while a fine-tuned BERT transformer achieved **71.43%** on the identical task, a gap too wide to close through feature engineering. The pattern recurs in the Dravidian setting, where classical classifiers on code-mixed text ranged only from **"inadequate to average,"** performing acceptably on majority classes but collapsing on the minority polarities that matter most (Chakravarthi et al., 2022). Together these results establish classical ML as the weak floor of the architectural evolution: its failures trace to specific design choices (context-free features and no cross-lingual transfer) that the contextual, subword-based transformers examined in Section 2.3.2 were explicitly built to overcome. This is why the central comparison of RQ2 is framed not as classical-versus-modern, but as a contest *within* the transformer family: **multilingual transformers versus monolingual baselines**.

2.3.2 Encoder-Based Multilingual Transformers

The limitations of classical models were largely resolved by the shift to **encoder-based transformers** such as BERT and its optimised variant RoBERTa. Rather than processing text as isolated tokens, these architectures use **self-attention** to read the whole sequence at once, producing **contextual embeddings** in which a word's representation is dynamically conditioned on its neighbours (Devlin et al., 2019; Selvaraju et al., 2024). This directly repairs the context-blindness diagnosed in Section 2.3.1, allowing the model to weight negation, intensification, and the cumulative polarity of a clause.

For code-mixed text, the field moved further, to **multilingual** transformers such as **XLM-RoBERTa (XLM-R)**, pre-trained across more than 100 languages so that tokens from different languages are projected into a **shared cross-lingual embedding space** (Conneau et al., 2020). In principle this lets the model read an English–Malay transition as a coherent semantic relationship rather than an anomaly. However, this is precisely where **RQ2** lives, and the evidence is more contested than it first appears. In introducing XLM-R, Conneau et al. (2020) also identified the **"curse of multilinguality"**: because model capacity is finite, spreading it across many languages dilutes per-language quality, so **monolingual baselines can match or exceed multilingual models on single-language tasks**, even as multilingual models retain the edge on cross-lingual transfer. Multilingual superiority is therefore **task-dependent, not guaranteed** — which is exactly the relationship RQ2 sets out to test rather than assume.

This contingency sharpens further in the regional context. Using their IndoRobusta benchmark, Adilazuarda et al. (2022) showed that a multilingual model's robustness to code-mixing is governed by **pre-training corpus representation**: Indonesian–English mixing was handled markedly better than mixing with lower-resourced local languages, **including Malay**, despite the latter's linguistic proximity. The implication for this project is significant: XLM-R's advantage over monolingual baselines on **Malay–English** text is plausible but **empirically uncertain and under-explored**, which is the gap RQ2 addresses directly rather than inheriting as an assumption.

Recognising that even well-represented multilingual transformers stumble on the irregular phonetic orthography of Bahasa Rojak, the literature offers two competing **adaptation strategies**, each with distinct trade-offs. The first is **transliteration-aware fine-tuning**: Manukonda and Kodali (2025) applied Masked Language Modeling over original, Romanised, and transliterated text so that XLM-R learns to equate non-standard phonetic spellings with their root meanings, mitigating the **out-of-vocabulary (OOV)** problem on a Dravidian task structurally parallel to Malay phonetic typing. Its strength is generality across scripts, but it depends on the availability of transliteration resources. The second is **corpus-specific pretraining with token-level language identification**: Romadhona et al. (2022) built the first Bahasa Rojak corpus (BRCC) and a *Mixed XLM* model that predicts each token's language alongside sentiment, forcing explicit awareness of code-switch boundaries. Its strength is that it is trained on **authentic Manglish** rather than a parallel language, but it is bounded by the size and coverage of the available Bahasa Rojak corpus. That **both** adaptations outperform vanilla XLM-R points to a clear conclusion: it is **domain adaptation, not raw multilinguality, that unlocks code-mixed performance**.

Taken together, the evidence positions a **fine-tuned, regionally adapted XLM-R** as the strongest encoder candidate for Malaysian code-mixed sentiment, yet one whose superiority over **monolingual baselines** remains contingent and untested in the Malaysian setting. This both motivates the project's use of a fine-tuned XLM-R (as an annotation voter in RQ1 and the evaluated classifier in RQ2) and frames the generative alternative examined next in Section 2.3.3.

2.3.3 Generative Large Language Models (LLMs)

The most recent stage of the architectural evolution moves from task-specific **fine-tuned encoders** to **generative Large Language Models (LLMs)**. Whereas an encoder such as XLM-R must be retrained on labelled data for each task, generative LLMs rely on **in-context learning;** by conditioning on a declarative prompt and a handful of examples, they perform **zero-shot** and **few-shot** classification without parameter updates (Brown et al., 2020). For an

annotation task (RQ1), this is a substantive shift, because it removes the dependency on the very labelled corpora that are scarcest for code-mixed Malay.

The **case for** generative models on Bahasa Rojak rests on their reasoning over context rather than surface tokens. Because LLMs are not bound to a fixed task vocabulary, they can in principle interpret the loose contextual cues, sarcasm, and rapid switches that trigger **OOV failures** in classical models, inferring sentiment from meaning rather than lexical match (Negi et al., 2026). This capability has begun to be demonstrated regionally; Khalip et al. (2025) applied an LLM to noisy Malay sentiment classification specifically to absorb the orthographic variation that defeats lexicon-based tools (Section 2.1.2). Used as **automatic annotators**, LLMs can thus generate labels through structured prompting alone, and have been shown elsewhere to rival human crowd-workers on comparable tasks (Gilardi et al., 2023; Negi et al., 2026).

This promise, however, must be weighed against **three architectural liabilities** a balanced review cannot omit. First, and most decisively for this project, there is the low-resource penalty: as established in Section 2.2.1, even frontier LLMs underperform a fine-tuned transformer on under-resourced languages, trailing BERT baselines by 10–14% accuracy (Jadhav et al., 2025), so generative reasoning does not compensate for thin pre-training coverage of a language like Malay. Second, there is reproducibility: single-pass LLM inference is stochastic and prone to hallucinated labels, so identical inputs can yield different outputs (Niimi, 2024; Negi et al., 2026). Third, there is computational cost: frontier models are expensive to run at corpus scale, which motivates the use of **quantised, locally hosted** mid-sized models as a practical compromise (Niimi, 2024). The first two liabilities are precisely what the consensus and adjudication mechanisms in Section 2.2.3 are designed to contain; the contribution of this section is to locate their *source* in the generative architecture itself.

The architectural verdict therefore mirrors the annotation-tier analysis of Section 2.2.1, but now on a mechanistic basis. The generative tier supplies the **contextual reasoning** the encoder tier lacks, yet forfeits the **low-resource reliability** the encoder tier provides. Because these strengths are **complementary rather than substitutable**, neither architecture alone is sufficient for Malaysian code-mixed sentiment, which is the precise rationale for this project's **hybrid pipeline**, pairing a fine-tuned XLM-R with a quantised LLM under majority voting so that each compensates for the other's failure mode. Having established the candidate architectures, the review turns in Section 2.4 to how their outputs, and the resulting dataset, must be *evaluated*.

2.4 Benchmarking and Evaluation Frameworks in Code-Mixed NLP

2.4.1 Evaluation Metrics for Imbalanced Data and Human-Machine Agreement

Evaluating this project requires two distinct measurement frameworks, one for each research question: a measure of how accurately the final classifier predicts sentiment on skewed data (**RQ2**), and a separate measure of how faithfully the automated annotation agrees with human judgment (**RQ1**). Conflating these is a common error, because a metric that is appropriate for one is misleading for the other. The need for care is amplified by the **severe class imbalance** typical of regional corpora; Mohd Shamsuddin et al. (2024), for instance, reported a distribution of 78.23% neutral, 16.33% positive, and 5.44% negative.

For **classifier accuracy (RQ2)**, raw predictive accuracy is actively deceptive on such data, a pitfall known as the **Accuracy Paradox** (Sokolova & Lapalme, 2009). A trivial classifier that labels every input "neutral" would score 78.23% accuracy on the corpus above while possessing zero ability to detect the minority polarities that carry the actual signal. The standard corrective, now near-universal in code-mixed benchmarking, is the **Macro-F1 score**, which computes precision and recall per class and averages them **unweighted**, so that performance on rare positive and negative slang is given equal mathematical weight to the dominant neutral class (Sokolova & Lapalme, 2009; Chakravarthi et al., 2022). Macro-F1 is preferred over **micro-** or **weighted-averaging** precisely because the latter let the majority class dominate the score and conceal minority-class failure, the same skew the annotation pipeline is meant to correct (Mohd Shamsuddin et al., 2024).

For **annotation reliability (RQ1)**, F1-based metrics are the wrong instrument entirely: they measure prediction against a gold label, not whether the labelling *process* is itself trustworthy. The appropriate measure is **inter-rater reliability (IRR)**, quantified via **Cohen's Kappa** for two raters or **Fleiss' Kappa** for three or more, which express agreement after subtracting the agreement expected by chance (Cohen, 1960; Fleiss, 1971), with interpretation typically following the Landis and Koch (1977) thresholds. This is the natural metric for validating **human–machine alignment** between the automated ensemble and expert annotators.

However, Kappa carries a limitation that is acutely relevant to this project rather than incidental. Under **high class imbalance**, the defining condition of this corpus, Kappa is subject to the **prevalence paradox**: because a dominant class inflates the expected chance agreement, high raw agreement can collapse into a misleadingly low Kappa, so the statistic reflects class prevalence as much as annotator quality (Feinstein & Cicchetti, 1990). Reporting Kappa alone on 78%-neutral data would therefore risk *understating* genuine human–machine alignment. The defensible response is to report Kappa alongside **raw percent agreement and class**

prevalence, or to adopt a **prevalence-robust coefficient such as Gwet's AC1**, which is designed for exactly this high-agreement, imbalanced condition (Gwet, 2008).

Taken together, a sound evaluation framework for this project pairs **Macro-F1** for RQ2 with a **prevalence-aware agreement measure** for RQ1, ensuring that neither the neutral majority nor inflated chance agreement can artificially inflate the reported results. With these instruments defined, Section 2.4.2 turns to the existing regional corpora these metrics would be applied to.

2.4.2 Existing Regional Corpora Benchmarks

Because this project's first contribution (**RQ1**) is a dataset, its value depends on what existing regional corpora already provide and, more importantly, what they collectively fail to provide. High-quality, human-validated corpora for code-mixed Malay–English text remain critically scarce, and the available resources can be compared along four criteria that determine fitness for training a robust general-purpose classifier: **scale**, **source authenticity**, **annotation method**, and **polarity balance** (Table 2.2).

Table 2.2: Comparison of Existing Malay–English Code-Mixed Sentiment Corpora

Corpus (Year)	**Scale**	**Source and Domain**	**Annotation Method**	**Polarity Scheme and Balance**
MY-EN-CS — Kasmuri & Basiron (2019)	~2,316 sentences	Blogs; local topics	Manual, 2 annotators (κ = 0.83)	Subjectivity only (opinionated vs factual)
MyCovid-Senti — Kong et al. (2023)	11,568 tweets	Twitter; COVID-19 only	Manual	3-class polarity; pandemic-bound
MESocSentiment — Mohd Shamsuddin et al. (2024)	19,714 posts	Twitter; general (#Malaysia)	Semi-automated consensus	3-class; severe skew (78.2 / 16.3 / 5.4)
SentiBahasaRojak/BRCC — Romadhona et al. (2022)	Augmentation-expanded	Partly formal/financial platforms	Augmentation + annotation	Trilingual (MS–EN–ZH) polarity

Grouped by annotation strategy, the trade-off identified in Section 2.2.1 becomes concrete at the corpus level. The manually annotated resources (Kasmuri & Basiron, 2019; Kong et al., 2023) achieve high reliability, with Kasmuri and Basiron (2019) reporting a Cohen's Kappa of 0.83, but pay for it in scale and breadth: at roughly 2,300 sentences and 11,568 tweets respectively, and confined to blog topics or the pandemic, neither can generalise to everyday Manglish. The **semi-automated** MESocSentiment (Mohd Shamsuddin et al., 2024) inverts the trade-off, reaching general social-media scale but inheriting the **78.2% neutral skew** that its shallow voting stack produced, leaving its minority classes too thin to train on. The **augmentation-based** SentiBahasaRojak (Romadhona et al., 2022) is the closest in spirit to authentic Manglish and uniquely handles trilingual mixing, yet its partial reliance on **formal, financial-platform text** and synthetic augmentation means it under-represents the unstructured phonetic orthography of casual social media.

The decisive observation is what emerges only from the comparison: **no existing corpus simultaneously satisfies all four criteria**. Scale is achieved only by sacrificing balance or human validation; authenticity and reliability are achieved only at unscalably small sizes. There is, in short, no large-scale, polarity-balanced, casually authentic, *human-validated* Manglish sentiment corpus — precisely the resource RQ1 sets out to construct, and the data-scarcity half of the research gap formalised in Section 2.5.

2.5 Critical Analysis and Summary of Research Gaps

2.5.1 Literature Synthesis Matrix

The preceding sections evaluated linguistic typologies, annotation methods, encoder variants, and generative models largely in isolation. Table 2.3 consolidates the most representative works so that the **cross-cutting pattern** they share, invisible from any single study, becomes explicit.

Two trends emerge. First, there is an **inverse relationship between flexibility and reliability**, as systems progress from static lexical matchers, to fine-tuned multilingual encoders, to zero-shot generative LLMs, the dominant failure mode migrates from **out-of-vocabulary (OOV) parsing errors** toward **hallucination and output variance**. Second, and more consequentially, every approach occupies an **architectural silo**, each supplies one necessary capability (scalability, regional pattern recognition, or contextual reasoning) while lacking the others, and the consensus mechanisms attempted so far combine only **homogeneous** models. The matrix therefore localises the specific vacancy that a cross-architecture hybrid would fill, which Section 2.5.2 formalises as the research gap.

Table 2.3: Literature Synthesis Matrix of Prior NLP Approaches and Their Unfilled Gaps

Work	**Domain / Data**	**Architecture / Method**	**Strength → Unfilled Gap**
Mohd Shamsuddin et al. (2024)	Malay–English Twitter (MESocSentiment)	Semi-automated voting of lexical/rule tools (VADER, TextBlob, pysentimiento)	Scales annotation cheaply, **but** the voters share an OOV blind spot, yielding a 78% neutral skew and no contextual reasoning.
Romadhona et al. (2022)	Trilingual MS–EN–ZH (SentiBahasaRojak); *Mixed XLM*	Encoder with token-level language ID + data augmentation	Native Bahasa Rojak with boundary awareness, **but** partly formal-source and augmentation-bound, under-representing casual orthography.
Manukonda & Kodali (2025)	Transliterated Dravidian–English	Transliteration-aware XLM-R (Masked Language Modeling)	Resolves phonetic OOV via fine-tuning, **but** rigid parameter weights leave it without zero-shot reasoning for unseen slang.
Negi et al. (2026)	Fine-grained opinion (general domains)	Generative LLMs as automatic annotators (zero-shot)	Deep semantic reasoning without labelled data, **but** single-pass inference is hallucination-prone and unstable without adjudication.
Jadhav et al. (2025)	Low-resource Indic annotation	LLMs vs. fine-tuned BERT, head-to-head	Establishes that LLMs trail fine-tuned transformers by 10–14% on low-resource tasks → **neither architecture alone is sufficient**.
Niimi (2024)	Localised review sentiment	Quantised local LLM + multi-attempt majority voting	Voting stabilises LLM variance, **but** only within one model type — not cross-architecture, and untested on Malaysian code-switching.

Read top to bottom, the final column exposes the gap directly, rows three to five contrast the **fine-tuned encoders** (reliable pattern recognition, no reasoning) against **generative LLMs** (strong reasoning, weak on low-resource and unstable), while the two consensus attempts (rows one and six) aggregate only **architecturally similar** models. No prior work unites the encoder's reliability with the LLM's reasoning under a single adjudicated consensus.

2.5.2 Identification of Research Gaps

Synthesising the trends in Table 2.3 against the requirements of the two research questions exposes **three interlocking gaps** that this project is designed to close.

Gap 1 — The data gap: an unresolved fidelity–scalability trade-off (RQ1). Robust regional classifiers are bottlenecked not by model capacity but by the scarcity of suitable training data, and existing curation methods force a trade-off that neither side resolves. **Manual annotation** yields high-fidelity labels but is prohibitively costly and unscalable for social-media volumes (Kasmuri & Basiron, 2019; Kong et al., 2023), whereas **single-layer automated pipelines** scale only at the cost of accuracy, defaulting to "neutral" on ambiguous bilingual slang and producing the 78.2% neutral skew that corrupts the minority polarities a classifier most needs (Mohd Shamsuddin et al., 2024). As Section 2.4.2 established, no existing corpus is simultaneously large-scale, polarity-balanced, casually authentic, and human-validated.

Gap 2 — The architectural gap: complementary models deployed in silos (RQ1). The literature treats the two leading architecture families as mutually exclusive rather than complementary. Fine-tuned **multilingual encoders** resolve phonetic OOV and recognise regional patterns reliably but cannot reason about unseen slang (Manukonda & Kodali, 2025), while **generative LLMs** reason across linguistic boundaries yet are hallucination-prone and, decisively, trail fine-tuned transformers on low-resource languages (Negi et al., 2026; Jadhav et al., 2025). The consensus methods that might bridge them have so far aggregated only architecturally **homogeneous** voters, shallow lexical tools (Mohd Shamsuddin et al., 2024) or repeated sampling of a single LLM (Niimi, 2024), so the cross-architecture combination, and the **human-in-the-loop adjudication** needed to anchor its hardest cases, remains unbuilt.

Gap 3 — The evaluation gap: multilingual vs. monolingual is untested regionally (RQ2). Although multilingual transformers such as XLM-R are widely *assumed* superior for code-mixed text, this superiority is **contingent rather than guaranteed**, constrained by the curse of multilinguality (Conneau et al., 2020) and by the pre-training under-representation of Malay specifically (Adilazuarda et al., 2022). The relative accuracy of multilingual transformers against **monolingual baselines** on Malaysian code-mixed sentiment therefore lacks direct empirical evidence and remains an open question.

Addressing the gaps. This project targets all three. To close **Gaps 1 and 2**, it constructs a Manglish sentiment corpus through a **heterogeneous tri-model majority-consensus pipeline** that combines a fine-tuned multilingual transformer with two generative LLMs from separate developers and resolves their disagreements under consensus, backed by **human-in-the-loop validation** to suppress single-model hallucination and the neutral-label skew. To close **Gap 3**, it evaluates that fine-tuned multilingual transformer against monolingual baselines on the resulting **human-validated test set**, supplying the regional evidence RQ2 requires while avoiding circular evaluation on machine-generated labels. Together, these establish a balanced, gold-standard Bahasa Rojak corpus and a validated framework for sentiment detection in code-mixed Malaysian text.

2.6 Summary

This chapter examined why sentiment analysis of Malaysian social media is difficult, and what the existing literature does and does not resolve. Section 2.1 established that **fluid English–Malay code-switching**, sentence-final particles, and phonetic orthography systematically defeat monolingual and lexicon-based tools through out-of-vocabulary failures. Sections 2.2 and 2.4 then showed that current annotation methods cannot escape a **fidelity–scalability trade-off**: manual corpora are reliable but small and domain-narrow, while single-layer automated pipelines scale only by producing a severe neutral-label skew. Section 2.3 traced the architectural evolution to its present impasse, fine-tuned multilingual transformers offer regional reliability but no contextual reasoning, whereas generative LLMs reason well yet falter on low-resource languages, with the two families consistently deployed in isolation rather than combined.

From this synthesis, Section 2.5 identified **three gaps**: the absence of a large-scale, balanced, human-validated Manglish corpus (**RQ1**); the lack of a cross-architecture consensus pipeline uniting transformer reliability with LLM reasoning under human-in-the-loop adjudication (**RQ1**); and the missing regional evidence on **how multilingual transformers compare with** monolingual baselines on Malaysian code-mixed text (**RQ2**). Chapter 3 details the methodology designed to close them: a **heterogeneous tri-model majority-consensus pipeline** (XLM-RoBERTa with two generative LLMs) for corpus construction, and a controlled evaluation against monolingual baselines on a human-validated test set, assessed using Cohen's Kappa alongside raw agreement and Gwet's AC1 for annotation reliability and **Macro-F1** for predictive performance.

Chapter 3

Methodology

3 Methodology

This chapter sets out the methodology the project follows to answer the two research questions stated in Chapter 1 and confirmed by the review in Chapter 2. It describes the **Heterogeneous Ensemble Annotation Framework** the project proposes, specifies how each part of it is built, applied and evaluated, and states the weaknesses the design carries so that they can be measured rather than hidden.

The chapter follows the order in which the framework runs. Section 3.1 gives the overall architecture and shows how the parts fit together. Section 3.2 explains the choice of source corpus and the preprocessing decisions that follow from it. Section 3.3 introduces the three models that vote on each post, and defends the claim that they are genuinely different from one another. Section 3.4 presents **confidence-gated routing**, which is the main contribution this design claims. Section 3.5 covers the model that resolves the disagreements the voters cannot settle. Section 3.6 sets out the two-stage validation design that establishes the quality of the output. Section 3.7 explains the comparative evaluation that answers RQ2. Section 3.8 records the environment the work runs in and what the project pins and publishes, and Section 3.9 summarises the chapter and evaluates the design against its own limitations.

How this chapter is organised. Each section states an operation or a decision and then justifies it immediately, so that the reasoning and the specification are read together rather than separately.

3.1 The Proposed Heterogeneous Ensemble Annotation Framework

Chapter 2 ended with a specific finding. Among the Malay-English code-mixed sentiment corpora that already exist, not one is at the same time large in scale, balanced across the three sentiment classes, drawn from casual everyday writing, and validated by human annotators. The manually annotated corpora are reliable, but they are small and tied to a single topic. The semi-automated corpus reaches general social-media scale, but its labelling method placed 78.23% of its posts in the neutral class (Mohd Shamsuddin et al., 2024). That leaves too few positive and negative examples for a classifier to learn from.

Chapter 2 also showed that the obvious repair does not work. Majority voting across several models fails when the voters share the same weakness, because they then agree on the same wrong answers. The gap is therefore not a shortage of models but a shortage of trustworthy labelled data, and no existing labelling method closes it. That argument belongs to Chapter 2 and is not repeated here.

This section describes the framework the project proposes in response. It gives the shape of the whole design first, then the three kinds of data the design works with, and then the two commitments that explain why the later sections are arranged as they are.

3.1.1 Architecture Overview

The project proposes a Heterogeneous Ensemble Annotation Framework.

Three models of deliberately different design read each post and predict its sentiment, and each one also reports how confident it is. These three models are called the voters. They are a fine-tuned XLM-RoBERTa (Conneau et al., 2020), Llama 3.1 8B (Llama Team, 2024), and Qwen2.5 7B Instruct (Qwen Team, 2024). Section 3.3 explains why these three were chosen and what makes them different from one another.

A label is accepted automatically only when the voters agree and are confident. Every other case is passed to a separate model that examines the disagreement and decides. A sample of the finished labels is then relabelled, independently and blind, by two human annotators, and that sample is what the framework's output is measured against.

Figure 3.1(a) and Figure 3.1(b) show the complete framework. The description below walks through it once, in the order the stages run.

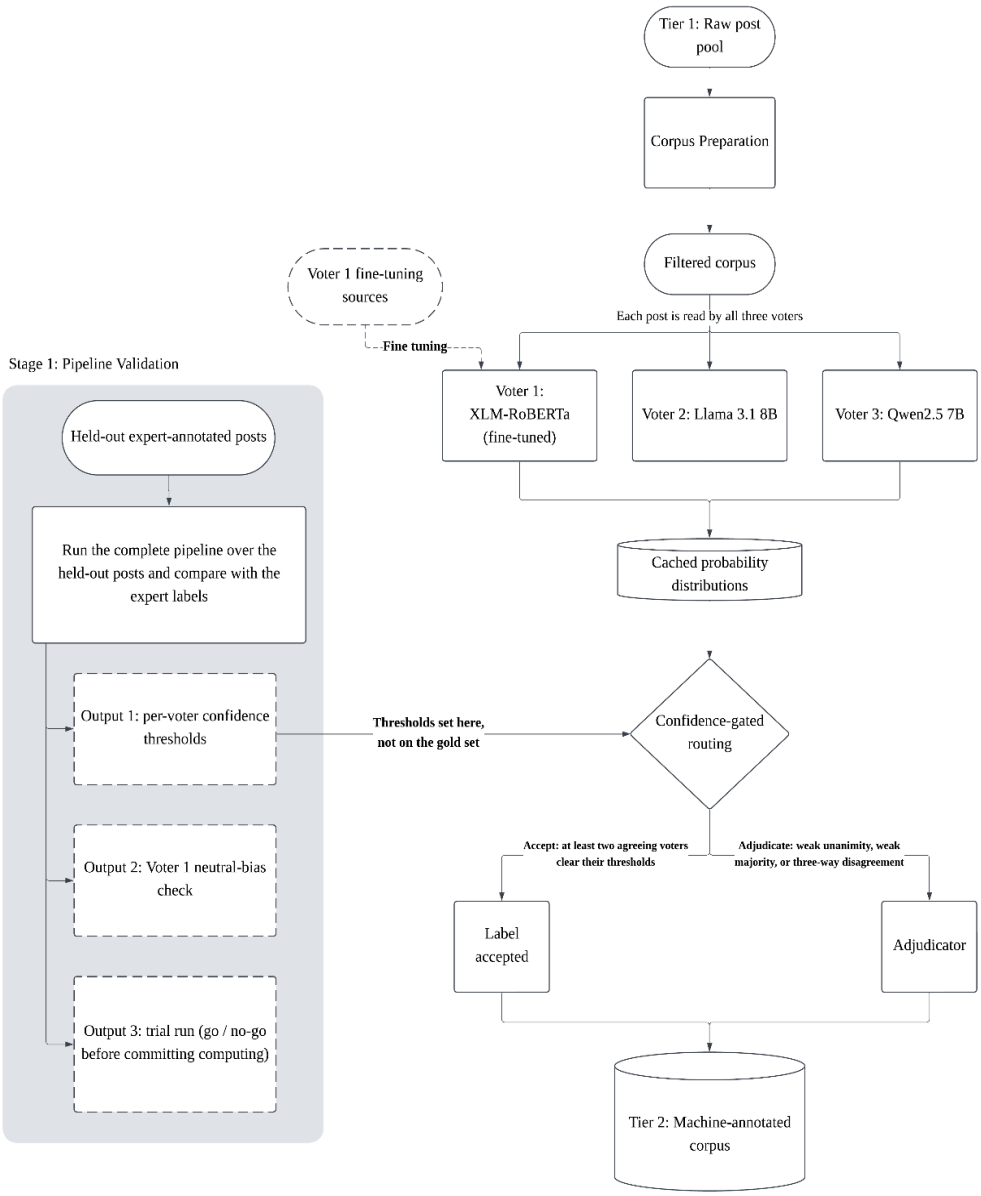


Figure 3.1(a): The annotation pipeline of the Heterogeneous Ensemble Annotation Framework

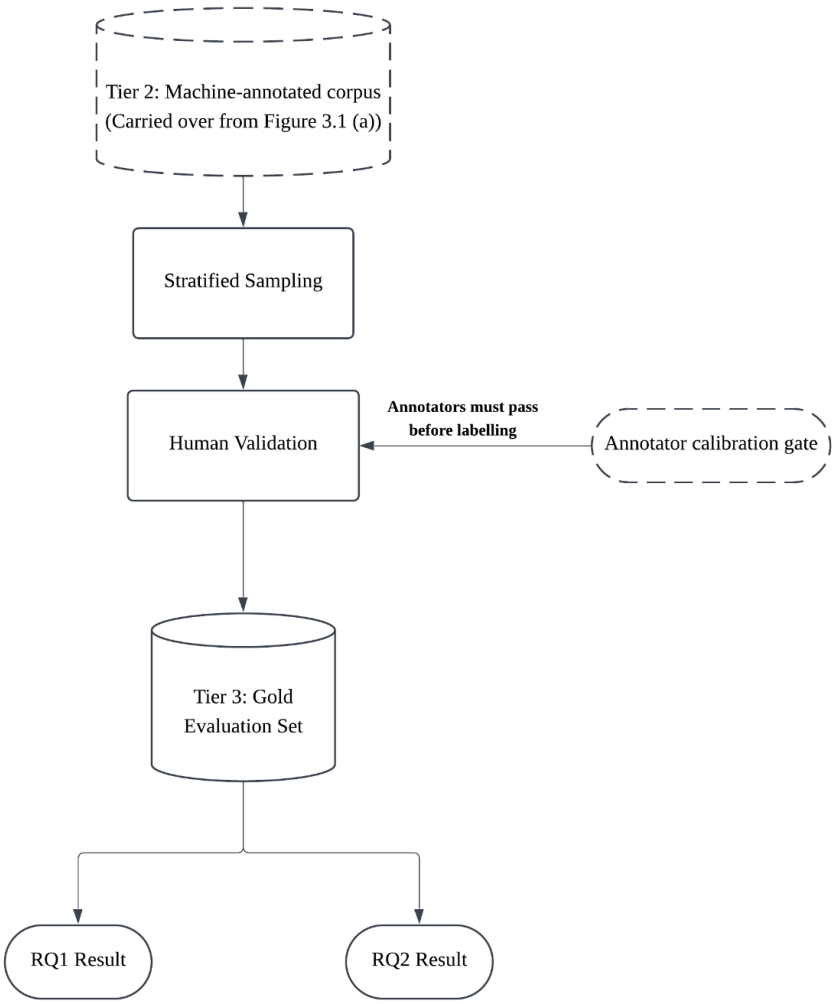


Figure 3.1(b): Validation and evaluation of the framework output

Corpus preparation. The raw post file is filtered to remove duplicates, very short posts, and spam or promotional material. A stable identifier is given to every surviving post, because the source file supplies none and the labels must later be joined back to it. A measure of how heavily each post mixes the two languages is also computed. This measure is not used to remove posts. It is recorded so that performance can later be reported separately for lightly and heavily mixed text.

Voting and routing. Every post is read by each of the three voters in turn, and each returns a predicted label together with a confidence score. A gate then examines the three votes and their confidence scores together. A label is accepted where at least two voters agree on it and both of those voters are confident. Every other post is passed on. This gate is the main contribution the design claims, and the confidence gate is set out in full in Section 3.4.

Adjudication. Posts passed on by the gate are decided by a different-lineage, regionally adapted reasoning adjudicator. This is a separate model, taken from a third model family, adapted to Southeast Asian languages, and run on local hardware rather than through a commercial service. It sees the post and the votes that conflicted, and it reasons through the disagreement instead of discarding the post. The labelled corpus is the output of these stages.

Validation and evaluation. Two validation stages sit around the annotation run. Before the run, the complete framework is applied to a set of posts that three expert annotators have already labelled. This is where the confidence thresholds are set, and where a poor design would show itself before a large amount of computing time is spent. After the run, a stratified sample of the finished corpus is relabelled by two bilingual annotators who cannot see the machine labels. The agreement between the two sets of labels is the evidence for H1. Finally, three sentiment classifiers, one multilingual and two monolingual, are trained on the same data and marked against that human-validated sample. That comparison is the evidence for H2.

3.1.2 The Three Tiers of Data

The framework works with data at three separate tiers, and keeping them distinct removes a confusion that otherwise runs through the whole design. All three tiers are **sets of posts, not models**. The models in this design are the three voters and the adjudicator described above.

Tier 1 is the raw pool. This is the collection of posts before any filtering, described in Section 3.2. It carries no sentiment labels, which is the reason the framework is needed at all.

Tier 2 is the machine-annotated corpus. These are the posts after the framework has labelled them, and this labelled corpus is the deliverable for RQ1. The project commits to annotating at least 150,000 posts and works towards the full filtered corpus. The lower figure is a floor rather than an estimate. It is set well below what the available hardware is expected to process, so that the contribution claimed for RQ1 does not depend on the annotation run going better than planned, and so that a shortfall would change the size of the result rather than the design. Even at the floor the corpus is more than ten times the size of the largest human-validated Malay-English code-mixed sentiment corpus reviewed in Chapter 2. Posts annotated above the floor are reported as a larger result rather than as a revised commitment. The final figure is fixed by measurement rather than by choice. A pilot run pushes between 5,000 and 10,000 posts through all three voters and records two quantities: the throughput of each voter, and the share of posts the gate sends to adjudication. Together these project the wall-clock time the full filtered corpus would take on the hardware described in Section 3.8. The annotation run is then set at the largest figure that projection supports within the project timetable, and never below the committed floor.

Tier 3 is the gold evaluation set. This is a stratified sample of at least 500 posts, and up to 1,000 where annotator time allows, drawn from Tier 2 and relabelled by two bilingual human annotators. It is not a product of the framework. It is the instrument used to measure the framework, and later the instrument used to mark the models in RQ2.

Tier 2 is the product. Tier 3 is the measuring instrument. The distinction matters because the project is easily misread as producing only 500 to 1,000 labels, which would be a modest contribution. That is not what it produces. It produces the Tier 2 corpus, and the role of the human annotation is to establish how far that corpus can be trusted.

The contribution is therefore the ratio between the two. Roughly a thousand human judgements are used to justify confidence in tens of thousands of machine labels. Manual annotation cannot reach the scale the field needs, and unchecked machine annotation cannot be trusted at any scale. Measuring the machine labels against a human-validated sample is what allows the project to claim both scale and trustworthiness at once.

3.1.3 Design Commitments and Traceability

Two decisions run through every later section, and stating them here explains why those sections are shaped as they are.

Heterogeneity is built in rather than incidental. Chapter 2 showed that a consensus of three models can still fail badly if the three fail in the same way. The voters in this framework are therefore chosen so that their differences are structural. One reads a post as a whole and classifies it directly, while the other two generate an answer as text. All three were pre-trained on different data by different teams, and all three split text into pieces using different methods. Section 3.3 defines this precisely and also states where the claim is weakest.

Nothing used to build or tune the framework is later used to judge it. The gold evaluation set is kept apart from every stage that could have shaped how the framework behaves. It is not used to train any voter, it is not used to set the confidence thresholds, and it is not used to train the models compared in RQ2. Section 3.6 sets this separation out in a table, because it is the clearest answer available to the objection that the project marks its own work.

Table 3.1 maps each research question to the objective it belongs to, the section that sets out the design, what that design produces, and how the result is measured.

Table 3.1: Mapping of research questions, objectives, design sections, outputs and metrics

Research question	Objective	Design set out in	Output	Measured by
Supporting	Objective 1	Section 3.2	Filtered corpus of code-mixed posts, ready for annotation	Posts retained after each filter
RQ1	Objective 2	Sections 3.3, 3.4 and 3.5	Tier 2 machine-annotated corpus	Share of posts resolved by each route, and the adjudication rate
RQ1	Objective 3	Section 3.6	Tier 3 gold evaluation set, and the evidence for H1	Cohen's Kappa with raw percentage agreement and class prevalence, or Gwet's AC1
RQ2	Objective 4	Section 3.7	Comparison of multilingual and monolingual classifiers, and the evidence for H2	Macro-F1 averaged across several random seeds, with robustness slices

The rest of the chapter takes each stage of Figure 3.1 in turn and explains why it is designed as it is.

3.2 Corpus and Preprocessing

This section sets out which body of text the framework annotates, and what is done to that text before the annotation begins. Two decisions carry weight for everything that follows: the choice of source corpus, and the decision to leave most of its surface features untouched.

3.2.1 Selection of Target Corpus

The framework annotates the X-post corpus released by Maskat et al. (2024). It was chosen mainly for a property that would normally be a shortcoming, which is that it carries no sentiment labels at all.

The corpus was built for a different purpose. Its authors annotated it for gender and for authorship, so that models could be trained to work out who wrote a post and whether the writer was male or female. Sentiment was never part of that work. The corpus therefore arrives with exactly what the framework needs: a large quantity of genuine Malaysian code-mixed writing,

and no existing sentiment labels that a new labelling method would be required to agree with or argue against.

That absence is the reason the corpus is suitable, not a defect to be worked around. Chapter 2 showed that the Malay-English corpora which do carry sentiment labels fall into two groups, and neither can be extended. The manually annotated corpora are reliable but small, and the effort that made them reliable is the same effort that stopped them growing. The large semi-automated corpus reached social-media scale, but its labels were shown to be unreliable, so building on it would mean building on top of the error the project exists to correct. A corpus with no labels avoids both problems, because there is nothing to inherit.

Two further properties matter. First, it is casual social-media writing rather than news, reviews or forum posts, so it contains the informal features Chapter 2 identified as sentiment-bearing in Manglish. Second, it is code-mixed throughout rather than a general corpus from which code-mixed posts must be extracted. One condition of use is still open: the terms under which the corpus may be redistributed are not yet confirmed, which is why Section 3.8 designs the release to carry labels alone rather than the posts themselves.

3.2.2 Corpus Composition and Its Limitations

The corpus contains 709,012 posts written by 50 authors, an average of about 14,180 posts per author, with the most prolific account contributing 47,067 posts and the least 2,300. The gender split across those accounts is 380,865 posts by female authors and 328,147 by male authors.

The author concentration follows from the corpus's original purpose. Authorship attribution needs many posts from each writer, not a few posts from many writers, so the sampling was deliberately author-centred. That design has a consequence the project must state rather than leave to be discovered, since the author count can be obtained from the released file in a single line of code. A corpus of 709,012 posts from 50 accounts is a large sample of fifty people's writing, not a cross-section of Malaysian social media.

The concentration creates three separate problems, each dealt with in a later section rather than here.

It weakens the scale claim. Chapter 2 compared existing regional corpora partly on size, and this corpus is far larger than any of them. That comparison remains valid, but it must be stated as scale in posts with acknowledged concentration in authors, not as scale alone. Anyone training on the released corpus inherits fifty people's vocabulary, topics and interests along with the labels.

It makes the gold sample vulnerable to a few accounts. Without a limit, a stratified sample of a thousand posts could be dominated by two or three prolific writers, and the human

validation would then measure how well the framework handles those writers rather than the corpus.

It leaves the RQ2 evaluation with only 50 units to split. An author-disjoint split between training and test data is the standard defence against a model learning writing style instead of sentiment, and at roughly 14,180 posts per author that defence is essential. But splitting 50 authors leaves a test set built on a handful of writers. Section 3.7.2 addresses this directly, including a second test set drawn from a corpus with far broader author coverage, so that the RQ2 result does not rest on this corpus alone.

None of these is solved by choosing a different corpus. No Malay-English code-mixed corpus exists at anything approaching this scale, so replacement is not an available option. The response is to measure the limitation rather than remove it.

3.2.3 Filtering Design

Because the corpus is code-mixed throughout, filtering has a narrower job here than it usually does. In most annotation projects the filtering stage decides which posts are worth keeping, and language identification does much of that work by separating the target language from everything else. Here there is nothing to separate. The consequence is that the size of the annotated corpus is limited by available computing time, not by data quality; every retained post is usable, and the only question is how many are processed above the floor committed to in Section 3.1.2.

Filtering is therefore kept light, and removes only posts that would waste computation or distort the result. Table 3.2 lists each filter and the reason for it.

Table 3.2: Filtering stages and their purpose

Stage	What it removes	Why it is needed
Duplicate removal	Exact repeats and near-duplicates	Reposts and copied text would be annotated repeatedly, inflating the corpus without adding new material, and would let identical posts fall on both sides of a train and test split
Length floor	Posts too short to carry a sentiment judgement	Very short posts give the voters almost nothing to read, so their labels would be close to arbitrary and would add noise rather than signal
Spam and promotional removal	Advertising, bulk marketing and automated posts	These are not personal expression, so a sentiment label for them is not meaningful, and their formulaic

Stage	What it removes	Why it is needed
		wording is over-represented relative to how often it occurs in real discourse

Two further operations run at this stage, and neither is a filter.

A stable row identifier is assigned to every post. The source file has three columns: Gender, Author and Xpost, and no identifier of any kind. Without one, the labels the framework produces cannot be reliably matched back to the posts they belong to. The identifier is created when the data is first read, and fixed from then on.

A code-mixing measure is computed for every post, recording how heavily it mixes the two languages. It describes posts rather than removing them. No post is removed on the basis of it.

Author identifiers are retained throughout, because the RQ2 split must be author-disjoint and cannot be built without them.

Finally, the assumption that the corpus is code-mixed throughout is checked rather than taken on trust. The expert-annotated corpus introduced in Section 3.3.2 marks each of its posts as Malay, English or code-mixed, giving an independent reference against which the project's own code-mixing measure can be compared. One caution applies: those language labels were assigned by a single annotator, while the sentiment labels in the same corpus were assigned by three. They are adequate as a check that the measure behaves sensibly, and are not treated as strong evidence about how common code-mixing is.

3.2.4 The Normalisation Decision

This subsection contains the most consequential preprocessing decision in the chapter, and it is a decision to do less rather than more.

The standard preparation applied to social media text is to convert it to lower case, strip punctuation, and remove emoji. It is applied so routinely that it usually passes without being mentioned. Chapter 2 argued that in Manglish this recipe removes the sentiment rather than the noise because emoji frequently carry the only explicit sentiment marker in a short post, capital letters and repeated punctuation mark emphasis, and elongated spellings such as "lahhh" or "ramaiii" signal intensity. A pipeline that strips these features deletes the signals the project is trying to detect.

The corpus makes this unusually easy to demonstrate, because its authors released two versions of the same data: a raw file, and a pre-processed file to which that standard recipe has already

been applied. Comparing them shows the effect directly rather than as a claim about what would happen. Table 3.3 shows two posts in both versions.

Table 3.3: The same posts before and after the corpus's supplied preprocessing

Post	Raw Version	Pre-process Version
1	Baru nak active jadi seller kat TT tapi mcm ramai orang hilang duit kat sana.. cm takowdt pulak 😔	baru nak active jadi seller kat tt tapi mcm ramai orang hilang duit kat sana cm takowdt pulak
2	Takpelah. After jogging maybe akan rasa better balik 🤪	takpelah after jogging maybe akan rasa better balik

In the first post the words report a situation where people are losing money on that platform and the sad emoji is what marks the writer's attitude towards it. Once it is removed, what remains reads much closer to a factual statement. In the second, the words are resigned and the emoji is what makes the post positive; without it the post reads flat. In both cases the transformation has not cleaned the post; it has removed the part that carries the sentiment.

Three measurements confirm the damage is not limited to a few examples. The raw file contains 3,369 duplicate posts; after the supplied preprocessing that figure rises to **52,037**, meaning stripping surface features collapsed tens of thousands of distinct posts into strings identical to some other post. The pre-processed file also contains 587 rows with no text at all, against none in the raw file. And it contains 650,409 rows against the raw file's 709,012, so **58,603 posts are absent from it entirely**, with no record of which ones or why. Because posts were dropped without an identifier being assigned first, the two files cannot even be reliably matched to each other by position.

The project therefore uses the raw file, and this is a defended decision rather than a default. The features a standard pipeline would remove are retained deliberately:

- **Emoji**, because in short posts they are often the clearest and sometimes the only explicit indication of the writer's attitude.
- **Capitalisation**, because capitals are used for emphasis rather than only at the start of sentences.
- **Elongation and repeated punctuation**, because both mark intensity: how strongly something is felt, rather than whether the feeling is positive or negative.
- **Phonetic and shortened spellings**, such as the Malay negator appearing as tak, tk or x, because these are the ordinary written forms in this variety rather than errors in it.

> Normalising them to dictionary Malay would train and evaluate the models on a form of the language Malaysians do not actually write.

A few light operations are still applied, because they remove material that carries no sentiment and would otherwise be learned as a signal. Web links and user mentions are handled here, because neither expresses the writer's attitude, and user mentions are tied to individual accounts, so leaving them in would give the models an easy route to identifying the author instead of the sentiment. Surrounding whitespace is tidied. Nothing else is changed.

The decision carries a real cost. Retaining the variation means the models must cope with a far wider range of surface forms for the same word, which is harder than being handed a normalised version. That is accepted deliberately, because Chapter 2's argument was that models unable to tolerate this variation are exactly the models that fail on Malaysian code-mixed text. Removing it before testing would produce a result about a language nobody writes in.

3.3 Heterogeneous Voter Panel

The voter panel is the part of the framework that produces the labels. Three models read each post independently, and each returns a predicted sentiment together with a confidence score. This section explains what makes the three different from one another, why that difference is the point of the design rather than a detail of its implementation, and where the claim is weakest.

3.3.1 Defining Heterogeneity and Its Limits

Chapter 2 established that combining several models does not by itself improve labels. The clearest evidence is MESocSentiment, the corpus that motivated this project (Mohd Shamsuddin et al., 2024). Its labels were produced by a majority vote across three structurally different tools: a rule-based analyser, a lexicon-based analyser, and a pre-trained transformer, but the result was still 78.23% neutral.

The reason is visible in the corpus authors' own reporting. Their Table 8 gives each analyser's output separately: of 19,714 posts, TextBlob assigned neutral to 14,305, VADER to 13,115, and pysentimiento to 16,916, which is between 67% and 86% each (Mohd Shamsuddin et al., 2024). The three did not disagree and then get averaged into a neutral majority. They agreed, because all three depended on matching words against a stored vocabulary, and Malaysian code-mixed text is full of words that none of those vocabularies contain. A majority vote can only cancel an error that some voters make and others do not. Here there was almost nothing to cancel.

The lesson is that **different implementation is not the same as different failure**. Three tools built in three different ways still share a blind spot if all three depend on the same thing.

Heterogeneity is therefore only useful when it is defined by properties that predict independent failure, and this framework defines it by three.

Different architecture, and a different route to the label. Voter 1 (XLM-RoBERTa) is an encoder: it reads the whole post at once and produces a score for each of the three classes directly. Voter 2 (Llama 3.1 8B) and Voter 3 (Qwen2.5 7B Instruct) are decoders: they generate the answer as text. These are two genuinely different ways of arriving at a judgement, not two settings of the same method.

Different pre-training lineage. The three models were built by different teams, at different times, on different collections of text. A weakness that comes from something missing in one model's pre-training data is unlikely to be shared by a model trained on different data.

Different tokenisation. Each model splits text into pieces using its own learned vocabulary. This matters specifically for Manglish. Chapter 2 showed that unfamiliar phonetic spellings are broken into fragments by subword tokenisers, and where those breaks fall depends on the vocabulary doing the splitting. Two models with different vocabularies fragment the same unusual word in different places, so they are unlikely to lose the same word in the same way. Table 3.4 sets out the panel against these properties.

Table 3.4: The voter panel and the properties that make it heterogeneous

Property	**Voter 1**	**Voter 2**	**Voter 3**
Model	XLM-RoBERTa (fine-tuned)	Llama 3.1 8B	Qwen2.5 7B Instruct
Architecture	Encoder	Decoder	Decoder
How the label is produced	Classified directly from the whole post	Generated as text	Generated as text
Pre-training lineage	Separate	Separate	Separate
Tokenisation	Own vocabulary	Own vocabulary	Own vocabulary
Role in the panel	Regional pattern recognition after fine-tuning	Contextual reasoning	Contextual reasoning
Source of confidence	Probability across the three classes	Probability of the label token	Probability of the label token

Where the claim is weakest. Two of the three voters are generative decoder models. That is a weaker separation than three unrelated designs would give. If a weakness is characteristic of instruction-tuned generative models in general, then Voters 2 and 3 could agree with each other and both be wrong. Malay sarcasm is the clearest candidate: the literal wording points one way and the speaker's intent the other, and a model reading the words closely can be misled by exactly the wording a Malaysian reader would recognise as mockery.

Three parts of the design respond to this, and none of them is a claim that it will not happen.

- Section 3.4 does not accept any agreement on agreement alone, including unanimous agreement. An agreement held with low confidence is routed away rather than accepted.
- Section 3.5 sends those cases to an adjudicator drawn from a third lineage, so a shared decoder bias is not the last word on them.
- Section 3.6.1 measures the panel against expert labels before the full run, which is where a shared bias of this kind becomes visible instead of staying hidden inside the output.

The expert-annotated corpus used for that validation carries sarcasm labels alongside its sentiment labels, so whether agreement between the voters degrades on sarcastic posts can be measured directly rather than assumed.

3.3.2 Voter 1: The Fine-tuned Multilingual Encoder

Voter 1 is XLM-RoBERTa, fine-tuned on Malaysian code-mixed sentiment data.

It is an encoder for two reasons. It reads the post as a whole rather than generating an answer, which is the complement of what the other two voters do; and it produces a probability across the three classes directly. It is fine-tuned because Chapter 2 showed that XLM-R's pre-training under-represents Malay, so it arrives with broad multilingual competence but limited exposure to the specific variety this project works on.

The objection this raises. Chapter 2 criticised the MESocSentiment corpus by name for its neutral skew. Fine-tuning a voter on that same corpus looks like a contradiction. The answer is not that the criticism was overstated. It is that the corpus's errors have a measurable structure, and that a decision about training data should rest on that measurement rather than on assumption.

The measurement is available because the corpus ships a 2,000-post test set carrying both its automatic labels and human labels for the same posts. Comparing the two columns measures the corpus's label quality directly. Raw agreement is 66.15% and Cohen's Kappa is 0.192, which

is "slight" agreement on the Landis and Koch (1977) scale. Table 3.5 gives the breakdown by class.

Table 3.5: Agreement between MESocSentiment's automatic labels and the manual labels shipped for the same 2,000 posts

Class	Recall	Precision	Labelled automatically	Labelled by a human
Negative	0.036	0.359	39	388
Neutral	0.867	0.747	1,609	1,386
Positive	0.473	0.304	352	226

This is the project's own analysis of the released files rather than a figure reported by the corpus authors. They acknowledge in their paper that incorrect labels are present and that the imbalance is a limitation of their approach, but they do not report an agreement statistic, so none is attributed to them. The calculation uses both label columns of the released 2,000-post test set and can be reproduced by anyone holding the public corpus.

What the numbers show. The dominant failure is misfiling into neutral. Of 388 posts a human marked negative, only 14 were labelled negative, while 291 went to neutral and 83 to positive. The automatic labels recover under four percent of genuine negatives. This is the catch-all behaviour Chapter 2 described.

The point that matters for this section, however, is that the damage is not confined to the neutral class. Precision on both minority classes is also low: of the 39 posts labelled negative, 14 actually were, and of the 352 labelled positive, 107 were. The corpus therefore cannot be treated as a dependable source of labels for any of the three classes, including the two Chapter 2 criticised less directly.

That has a direct consequence for the design, and it is why the training data for Voter 1 is not simply asserted. The expert-annotated public-security corpus (Md Suhaimin et al., 2024a, 2024b) is the primary source, because its sentiment labels were assigned by three expert annotators and its class distribution is far healthier, at 4,197 negative, 2,569 neutral and 3,072 positive across 10,000 posts, of which 162 reached no majority and carry no label. It carries two limitations of its own: it is drawn from one narrow domain, and only 2,320 of its rows are code-mixed. MESocSentiment offers what that corpus lacks, which is general social-media coverage and volume, but the measurement above shows that it cannot supply the label authority to go with them.

Neither source is therefore adequate on its own, and whether MESocSentiment is used at all, and in what proportion, is settled by a controlled comparison rather than by a judgement call. Three conditions are compared: the general social-media corpus alone, the expert-annotated corpus alone, and the two combined with the majority neutral class subsampled so that the combined set is not dominated by it. Each condition is trained under three random seeds and scored by Macro-F1 against the held-out expert portion described in Section 3.6.1, and the mean across the three seeds decides the condition. The hyperparameters are held identical across the three conditions, so that the training data is the only thing that varies. The comparison runs to completion before any annotation begins, because the voter that helps produce the corpus cannot be chosen after the corpus exists.

One further caution shapes how that comparison is built. MESocSentiment ships a pre-balanced training split, and its provenance does not hold up on inspection. It contains 10,863 rows but only 9,906 distinct posts, the balance having been reached by repeating minority-class posts rather than by reducing the majority class. Only 55.3% of its posts appear in the released corpus at all, and where they do appear the labels disagree in 17.4% of cases, including 497 posts the corpus calls neutral and the training split calls negative. The project therefore builds its own splits from the released corpus rather than adopting the shipped one.

The framing that makes an imperfect Voter 1 acceptable. Fine-tuning here serves **linguistic adaptation, not label authority**. Voter 1's job is to have seen Malaysian code-mixed writing before, not to be the final word on any post. It holds one vote of three, its errors can be outvoted, and its low-confidence judgements are routed to adjudication. A somewhat imperfect Voter 1 is survivable. A quietly dominant one is not, and that is why the project checks its predicted class distribution against expert labels before the full annotation run, rather than discovering an inherited skew afterwards.

3.3.3 Voters 2 and 3: The Generative Voters

Voters 2 and 3 are Llama 3.1 8B and Qwen2.5 7B Instruct. Both are fixed choices rather than candidates still under consideration. If either cannot be run within the available graphics memory at **four-bit quantisation**, its replacement is a smaller release from the same developer rather than a model from another family, so that the lineage separation the panel depends on survives the substitution.

Why generative models are in the panel at all. Chapter 2 drew a consistent distinction between the two model families. Fine-tuned encoders recognise regional patterns reliably but do not reason about what they have not seen, while generative models reason about context, such as an unfamiliar slang term, a sarcastic construction, or a switch in the middle of a sentence, yet they perform worse than fine-tuned encoders on low-resource languages. These

strengths are complementary rather than competing, and pairing them is the reason the panel is built this way instead of from three encoders.

Why these two models. They come from different developers and different pre-training lineages and use different tokenisers. Their selection is justified against recent regional evaluation results rather than general leaderboards, because a model's average performance across many languages says little about its competence in Malay specifically. SEA-HELM (Susanto et al., 2025) is the evaluation used, because it scores Llama-3.1-8B-Instruct, Qwen2.5-7B-Instruct and the Gemma 2 9B lineage the adjudicator is drawn from on Southeast Asian languages at the same parameter sizes this project runs, and BHASA (Leong et al., 2023) covers the same region with a sentiment task. On SEA-HELM's Indonesian tasks, which include the NusaX sentiment dataset, Qwen2.5 7B Instruct scores 61.1 and Llama 3.1 8B 53.4 on a normalised scale to 100 (Susanto et al., 2025, Table 2). The more informative figure is how that difference moves between languages: the two models are 15.9 points apart on Thai, 7.7 on Indonesian and 0.4 on Filipino, and both fall to around 14 on Tamil. Two models of the same order of size whose relative standing swings that widely are not failing on the same inputs, which is the property Section 3.3.1 defines heterogeneity by, here measured rather than argued. Where evaluations covering Malay are unavailable, Indonesian results are used as the closest available proxy, and that substitution is stated rather than passed over: the two languages are closely related but not the same, so a result on one is evidence about the other rather than a measurement of it.

Why not one model queried several times. Chapter 2 covered the alternative of querying a single model repeatedly and taking its most frequent answer. That makes one model's output more stable, but it does not supply a second opinion, because a model's blind spot is still there on the fifth attempt. Repeating one voter and consulting three different ones are different operations, and only the second addresses the failure Chapter 2 identified.

Sequential execution. The available graphics card holds one generative voter in memory at a time, so the voters are run over the corpus one after another rather than together. This follows from the hardware rather than from a design preference, and its effect falls on the length of the annotation run, not on the labels. Each voter reads each post independently in any case, and there is no exchange between them that running them at the same time would enable.

Prompting and decoding. Both generative voters receive the same fixed instruction, and both are constrained to answer with one of the three labels instead of free text. Holding the prompt identical across the two means any difference in their answers comes from the models themselves, not from how they were asked. Both run at four-bit quantisation and decode greedily, so the same post under the same prompt returns the same label on every run. The

probability assigned to each of the three label words at the point of answering is read and renormalised, as Section 3.4.2 describes.

3.4 Confidence Extraction and Gated Routing

This section presents the mechanism the project claims as its contribution. Everything before it is preparation: the three voters and the differences between them exist so that this stage has something useful to work with. What the framework does with the three votes is where the design departs from the consensus frameworks Chapter 2 reviewed.

3.4.1 Why Agreement Alone Is Insufficient

The obvious way to combine three votes is a majority rule: accept whichever label two or more voters agree on. It is the rule the corpus criticised in Chapter 2 used, and it fails in a specific and predictable way.

The failure is that **agreement is not evidence of correctness**. Two models can agree because both read the sentiment correctly, or because both missed it in the same way. A majority rule cannot tell the two situations apart. It sees two identical votes in each case and accepts.

Section 3.3.1 showed that this is not a hypothetical concern. The three analysers that produced the earlier corpus agreed on neutral for most of it, because all three failed on the same inputs for the same reason, and the majority rule accepted every one of those agreements.

Two consequences follow. The first is that a majority rule does not merely pass shared error through, it **certifies it**. The output arrives looking like consensus, and consensus is what a reader treats as a sign of reliability, so the error is delivered with a mark of quality attached to it.

The second is less obvious and is what this framework acts on. **A majority rule treats every majority as equal.** A post where two voters were nearly certain and a post where two voters barely preferred one label over another produce exactly the same output. The second is far more likely to be wrong, and the rule has discarded the very information that would have revealed it because the voters do produce that information. A model that predicts a class also produces some measure of how strongly it prefers that class to the alternatives. A plain majority rule throws this away at the moment of voting, and then has nothing left with which to separate a firm agreement from a weak one.

Keeping that information, and requiring an agreement to be confident before it is accepted, is what this section adds.

One alternative deserves stating, because it is the natural first thought and it is not what this framework does. Confidence could be used to **weight** the votes, so that a confident voter counts for more than a hesitant one. Weighting still produces a label for every post, however confident

or not the panel was. The design here uses confidence to **route** rather than to weight: a weak agreement is not resolved by arithmetic; it is passed to something better equipped to resolve it. The framework is allowed to say that the panel did not settle a post, which a weighting scheme never says.

3.4.2 Extracting Confidence from Each Voter

Confidence has to be obtained from each voter in a way that suits how that voter produces its answer, and the two kinds of voter differ.

Voter 1 is an encoder. It produces a score for each of the three classes and converts them into probabilities that add up to one, so the probability of the class it chose is its confidence. Nothing extra is required.

Voters 2 and 3 are generative and produce text rather than a distribution over classes. The method used here constrains their answer to the three label words, and reads the probability the model assigned to each of those words at the point it answered, renormalised across the three. This gives a distribution of the same form as Voter 1's, at one pass per post.

The alternative is to ask each generative voter the same question several times and use how often it gives each answer as a measure of confidence. Chapter 2 covered this approach, and it works. It also multiplies the cost of the annotation run several times over, and at the size of this corpus that difference decides whether the run fits within the project timetable. Reading the probabilities directly obtains the same information in a single pass.

One method is rejected outright: **asking a model to state its own confidence in words**, for instance on a scale from zero to ten. It is easy to implement, which is why it is common, and it is not usable here. A figure a model writes as text is produced the same way as any other text it writes, so it reflects how such statements are usually phrased rather than how likely that particular answer is to be correct. Since the gate is the mechanism the framework's contribution rests on, it cannot be built on a signal with no dependable relationship to correctness. Table 3.6 sets out how confidence is obtained from each voter.

Table 3.6: How confidence is obtained from each voter

Voter	How confidence is obtained	Why this method
Voter 1 (encoder)	The probability assigned to the predicted class, from the distribution the model produces across all three classes	The model produces this directly, so no extra step is needed

Voter	How confidence is obtained	Why this method
Voters 2 and 3 (generative)	The answer is constrained to the three label words, and the probability given to each is read at the point of answering, then renormalised across the three	Produces a distribution of the same form as Voter 1's at one pass per post
Rejected for all voters	Asking the model to state a confidence figure in words	A figure written as text is generated like any other text and bears no dependable relationship to whether the answer is right

3.4.3 The Confidence-Gated Routing Rule

The rule takes the three votes together with their confidence scores and sends each post down one of two paths: accepted as labelled, or passed to adjudication. No post is discarded. The rule is stated first as a single condition, then justified, and Figure 3.2 shows it.

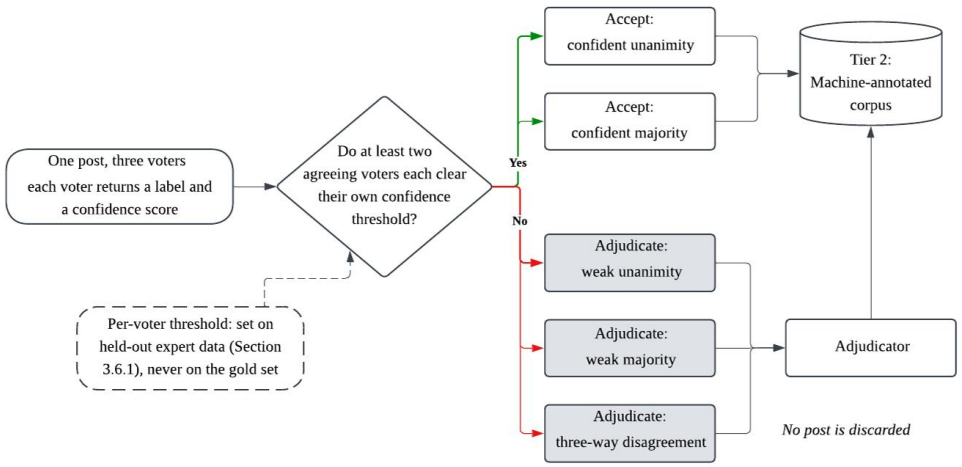


Figure 3.2: The confidence-gated routing rule

The rule. A label is accepted when at least two voters have predicted it and each of those two voters has cleared its own confidence threshold. Every other post is adjudicated.

With three voters and three classes, at most one label can carry more than one vote, so there is never a question of which label the condition is being applied to. Table 3.7 sets out every configuration the panel can produce, and where each one is sent.

Table 3.7: Every configuration the panel can produce, and the route it takes

The votes	The agreeing voters' confidence	Route	Why
All three agree	All three clear their thresholds	Accepted	Three voters built on different architectures, different pre-training data and different tokenisers agree, and none of them is hesitant. This is the strongest evidence the panel is able to produce.
All three agree	Two clear, one falls below	Accepted	Two independent confident voters agree, and no voter argues for a different label.
All three agree	One clears, or none	Adjudicated	The label would rest on at most one confident voter, which is the single-model judgement this framework exists to replace.
Two agree	Both clear their thresholds	Accepted	Two independent confident voters agree. The dissenting voter is not consulted further.
Two agree	At least one falls below	Adjudicated	The label would rest on one confident voter while a second voter argues for a different label.
All three disagree	Not applicable	Adjudicated	No label carries two votes, so the condition cannot be met.

The same test applies to every route into the corpus. Section 3.4.1 argued that agreement is not evidence of correctness, and that a plain majority rule fails because it treats every majority as equal. An earlier version of this rule accepted unanimous agreement outright, without any confidence condition attached to it. That left one branch exempt from the chapter's own argument, and it was the branch where a shared error is most damaging: a wrong label that all three voters agree on arrives carrying the strongest available appearance of consensus. Section 3.3.1 showed that three tools failing together on the same inputs is not a hypothetical case but is what produced the corpus this project exists to correct. The condition now applies wherever a label enters the accepted corpus, and no vote pattern is exempt from it.

Why the condition asks for two confident voters and not three. The stricter alternative is to require every agreeing voter to clear its threshold, so that unanimity is accepted only when all three are confident. That rule produces an outcome it cannot defend. Take two voters that agree on a label and both clear their thresholds. If the third voter argues for a different label, the

stricter rule accepts the post. If the third voter instead agrees with the other two but scores a little below its threshold, the stricter rule adjudicates it. A voter moving from disagreement to agreement would therefore move a post from accepted to unresolved, which treats an additional agreeing vote as evidence against the label it supports. The second configuration contains everything the first contains, two independent confident voters agreeing on one label, and where the first has a voter arguing for something else the second has a third voter pointing the same way. It cannot be the weaker of the two, so it cannot be the one that is refused.

Two confident voters is the unit of evidence the panel was built to produce. Section 3.3.1 defines the panel's heterogeneity by architecture, pre-training lineage and tokenisation, and those properties were chosen because they predict independent failure. What justifies accepting a label is therefore two confident judgements that are unlikely to be wrong in the same way, which is the smallest configuration a single model's error cannot produce on its own. A third agreeing voter strengthens evidence of the same kind rather than supplying evidence of a different kind, so requiring it does not answer an objection that the first two have left open.

A threshold is a tuned value, not a boundary between right and wrong answers. Section 3.6.1 states that the thresholds are set on held-out expert-annotated data drawn from a different domain from the target corpus, and treats them as starting values whose effect is tested rather than as settled constants. Under the stricter rule, one borderline score, possibly produced by a threshold set slightly too high for that particular voter, would override two confident agreements from separate lineages. Requiring two of the three leaves the gate tolerant of a single mis-set threshold. That is the appropriate design given a limitation the chapter has already conceded rather than discovered.

What the rule costs, in both directions. The routing rule also decides how much work reaches Section 3.5, and the two comparisons are not symmetrical. Against the earlier version of this rule, the revision sends more posts to adjudication, because weakly held unanimous agreements now go there instead of being accepted automatically. That cost is accepted deliberately, since it is the price of applying the confidence condition without exception, and Section 3.5.2 already records that adjudication is a major stage of the annotation run rather than a small final step. Against the stricter alternative, the revision sends fewer, and the posts it withholds are precisely those where no voter argued for a different label. Section 3.5.2 describes the adjudicator's task as reasoning over a conflict, and a post carrying no conflict is not the material that stage was designed for. The adjudicator does not outrank the voters in scale, so it has no particular claim to improve on two confident agreeing voters where nothing disputes them.

The choice is measured rather than settled by argument. Caching every voter's full probability distribution means the accepted and adjudicated shares under any of these rules can

be recomputed from stored numbers, at the cost of re-adjudicating only the posts that change path. Stage 1 in Section 3.6.1 runs the complete pipeline over held-out expert-labelled posts, so the accuracy of each route can be compared against expert labels before the full annotation run begins. The stricter rule is therefore a decision this project can revisit on evidence, and the sensitivity check reports the output under both.

A refinement considered and not adopted. Where two voters agree confidently and the dissenting voter is itself highly confident, the disagreement could be treated as strong enough to force adjudication. It is not, for two reasons. Section 3.4.1's argument applies to the dissenter as well as to the majority: a single model's confidence is not evidence that its answer is correct, so one confident dissenting voter does not outweigh two confident agreeing voters drawn from separate lineages. And the refinement would add a second confidence test, with its own tuning problem, to a rule whose value is that it asks one question. How often the case arises, and how often the dissenting voter turned out to be right, are both recoverable from the cached distributions and are reported as error analysis in Section 3.7.3 rather than built into the gate.

What is stored at this stage. The gate reads the confidence distributions but does not consume them. For every post, the renormalised distribution across the three labels is written for each of the three voters, together with the post identifier, the three predicted labels, the route the gate assigned, and the condition that sent it there. Storing the distributions rather than only the chosen labels is what makes the threshold sensitivity check and the rule comparisons above possible without repeating inference: changing a threshold changes which posts cross the gate, and that can be recomputed from the stored numbers, leaving only the posts that change path to be adjudicated again. These records are released with the labels, as Section 3.8.3 describes.

3.5 Adjudication

The gate in Section 3.4 sends three kinds of post onward: those where two voters agreed but at least one held that agreement weakly, those where all three voters agreed but only one of them held that agreement with confidence, and those where all three voters disagreed. Together these are the cases the panel could not settle.

Adjudication here is not a tie-breaker in the ordinary sense. A tie-breaker chooses between options that are already on the table. The adjudicator is given the post and the conflict and asked to work out why the conflict arose, which is a different and harder task. Chapter 2 established the principle: disagreement is information to be reasoned over rather than noise to be averaged away or deleted (Negi et al., 2026).

3.5.1 Selection of the Adjudicator

Why a locally hosted model rather than a commercial service. The obvious candidate for the hardest posts in the corpus is a commercial model, since the largest and most capable models are reached that way. The project rules this out for three reasons, and treats the decision as a strength it claims rather than a restriction it has accepted.

Cost. Adjudication handles a substantial share of the corpus rather than a handful of edge cases, because the gate routes weakly held agreements, unanimous ones included, as well as outright disagreements. Paying for each call at that volume is beyond the project's means, and it would make the size of the finished corpus a function of the budget available rather than of the design.

Reproducibility. A hosted model can change without notice and without a version a reader can pin. A result produced against one cannot be reliably reproduced later, because the thing that produced it may no longer exist in the same form. An open-weight model pinned to a specific release does not carry this problem. Since RQ1 makes a claim about a labelling framework rather than about one run of it, this matters more here than it would in a study reporting a single measurement.

Accessibility. Anyone who wishes to check or extend the work can obtain the same weights and run them. The alternative restricts verification to those able to pay for it, which for a resource intended as a public contribution defeats its own purpose.

Taken together, these mean the whole pipeline can be regenerated from public data and open model weights. What began as a constraint on the design is the property that makes it reproducible by someone else.

Which local model. Two designs were available, and they resolve disagreement in different ways. Table 3.8 sets them against each other.

Table 3.8: The two adjudicator designs considered

Property	**A larger model from a voter's own family**	**A different-lineage, regionally adapted model (selected)**
Resolves disagreement through	Greater capacity than the voters	A different training lineage and stronger regional competence
Lineage overlap with the voters	Shares a family with one voter	Shares a family with neither voter

Property	**A larger model from a voter's own family**	**A different-lineage, regionally adapted model (selected)**
Malaysian-language competence	Not specifically adapted	Pre-trained with Southeast Asian language coverage
Fit within available graphics memory	Must be split between the graphics card and system memory, lengthening the adjudication run considerably	Runs entirely on the graphics card
Main weakness	Inherits the blind spots of the family it shares with a voter	Does not outrank the voters in scale

The selected design is a **different-lineage, regionally adapted reasoning adjudicator**: a model drawn from a third family, pre-trained with Southeast Asian language coverage, which differs from the voters in what it knows rather than in how large it is. The SEA-LION family (Ng et al., 2025) is used, at the release built on the Gemma 2 9B lineage and run at four-bit quantisation, which is the configuration that fits entirely within the available graphics memory. If that release proves unusable, the replacement is a smaller SEA-LION release rather than a model from a voter's family, because the third lineage is the justification for the stage. The release version is pinned and recorded in Table 3.14.

Three justifications support the choice, in order of strength.

A third separate lineage. This is the strongest reason, and it follows directly from the argument in Section 3.3.1. An adjudicator drawn from the same family as one of the voters would carry that family's blind spots into the stage meant to correct them. A post the panel failed to settle *because* a voter's lineage handles that kind of post badly would then be handed to a model with the same weakness, and the adjudication stage would add nothing on precisely the cases where it is most needed. A third lineage means the adjudicator's own errors are unlikely to be the errors that caused the disagreement.

Regional adaptation. The adjudicator receives the corpus's hardest material: the ambiguous, heavily code-mixed posts the panel could not agree on. A model pre-trained with Southeast Asian coverage is better placed on that material than a general-purpose model of similar size, and SEA-HELM measures that comparison directly: the Southeast Asian-adapted release scores 63.2 on the regional average against 60.0 for the same Gemma 2 9B model without the adaptation, and 65.1 against 62.7 on Indonesian (Susanto et al., 2025, Table 2). Both figures sit

well above either generative voter, which score 46.1 and 39.7 on the same regional average, so the stage gains regional competence without gaining scale over the models whose disagreement it resolves. The choice supports the Malaysian-context argument running through Chapter 2 rather than sitting apart from it.

Feasibility. The model runs within the available graphics memory without being split across the card and system memory. A substantially larger model would have to be, and adjudicating the routed share of the corpus would then take long enough to put the project timetable at risk. This is the weakest of the three justifications and is listed last deliberately: it explains why the choice is workable, not why it is right.

A claim the project gives up. An earlier version of this design specified an adjudicator larger than the voters, on the reasoning that greater capacity would settle what the panel could not. That is no longer the design, and the change is stated here rather than left for a reader to infer from the model sizes. **The adjudicator does not outrank the voters in scale.** Its justification is lineage and regional competence, not capacity.

Being direct about this matters, because the change could be read as a weakening of the project. It is not, because the framework's contribution was never adjudicator scale. The contribution is the routing rule in Section 3.4, which decides which posts reach adjudication at all, and that rule is untouched by the change. What the change does affect is how much can be claimed for the adjudication stage itself.

3.5.2 Adjudication Procedure and Its Limitation

What the adjudicator is given. It sees the post, and it sees the votes that conflicted. It does not see which model cast which vote: the votes are presented without identifying their source, so the adjudicator cannot defer to whichever voter it might otherwise treat as the more authoritative. The prompt directs its attention to the features that make these posts hard, such as sarcasm, a switch in the middle of a sentence, or an unfamiliar term, and requires one of the three sentiment labels as the answer. The label is the whole of the output. It is not asked to justify its choice, and no account of its decision is stored or read. The project therefore makes no claim about interpreting why a particular label was returned. That boundary is deliberate rather than incidental: explaining a model's decisions is a separate research problem with its own methods and its own evaluation, and this framework does not enter it. What the design claims here is the instruction rather than the outcome. Whether the adjudicator succeeds on these cases is measured against expert labels in Section 3.6.1 and reported by route, rather than assumed.

Why it is shown the votes at all, rather than simply asked to label the post afresh. An independent fourth opinion would make the adjudicator a fourth voter, and would leave the

framework needing a rule to combine four votes where it previously needed one for three. Showing the conflict makes the task different in kind: the adjudicator is told where the difficulty lies and asked to resolve that specific difficulty. This follows the pattern Chapter 2 identified in the adjudication literature, where the value of the stage comes from reasoning over the disagreement rather than from adding another vote to it.

Nothing is discarded. Every post that reaches this stage leaves it with a label. This is the direct repair of the failure Chapter 2 documented in the baseline framework, which had no mechanism for genuine disagreement and removed those posts from its corpus, and that lost exactly the ambiguous, heavily mixed material that makes a corpus valuable for this task in the first place.

The volume this stage carries. The gate routes every weakly held agreement, unanimous ones included, as well as three-way disagreements, so the share of the corpus reaching adjudication is much larger than the rate of outright ties. This is the mechanism working as intended, not a problem to be reduced, but it means adjudication is a major stage of the annotation run and the compute budget in Section 3.8 is built accordingly.

The limitation. The adjudicator does not outrank the voters in scale, so there is no guarantee that it can settle what three voters could not, and the design does not claim one. A post routed here because it is genuinely difficult may be difficult for the adjudicator as well.

The response to this is measurement rather than further argument. Section 3.6.1 runs the complete pipeline over expert-annotated data held back for the purpose, and the cases routed to adjudication within that run can be compared against the expert labels separately from the rest. That yields a figure for **how accurate adjudication is on the cases actually sent to it**, reported as a distinct result rather than absorbed into the framework's overall agreement figure. If adjudication performs poorly on those cases, the design has a measured weakness rather than an unexamined assumption, and an alternative is recorded here rather than left open. Should the measurement call for one, the replacement is a larger model from the same third lineage, run at a more aggressive quantisation and a reduced context length to fit the same graphics memory. The substitution changes the capacity of the stage and not its justification, because the lineage separation is the reason the stage exists and must survive any replacement.

Reporting the figure separately has a second use. Because the routing rule distinguishes weak majorities from three-way disagreements, accuracy can be reported for each route independently. That shows not only whether adjudication handles the hard cases well, but whether the gate is sending it the right ones.

The adjudicator is constrained to the three label words in the same way as the generative voters, so an answer outside the label set cannot arise by construction; a post whose adjudication fails for a runtime reason is retried rather than dropped, which is what keeps the commitment that

no post is discarded. Three fields are written for every adjudicated post: the post identifier, the votes that conflicted, and the label returned. The pinned release identifier is recorded in Table 3.14.

3.6 Validation Procedure

The framework produces labels. This section sets out how the project establishes that those labels are worth having, and how it does so without measuring its own work using an instrument it built for the purpose.

Validation runs in two stages, at different points in the project and for different reasons. Stage 1 runs **before** the main annotation, over data the project did not label. Stage 2 runs **after** it, over the framework's own output. They are not two attempts at the same check: the first asks whether the framework is fit to run, and the second asks how good its output turned out to be.

3.6.1 Stage 1: Pipeline Validation on Held-out Expert Data

Before the framework is run over the target corpus, it is run over a set of posts that already carry expert labels. These come from the code-mixed portion of the public-security corpus introduced in Section 3.3.2, whose sentiment labels were assigned by three expert annotators. A portion of it is held back from the beginning and used for nothing else. In particular, it is never used to fine-tune Voter 1.

Running the complete pipeline over that portion (all three voters, the gate, and the adjudicator) and comparing the result against the expert labels does three separate jobs.

It sets the confidence thresholds, and it is the reason those thresholds can be trusted. Section 3.4.3 established that per-voter thresholds have to be tuned rather than guessed. Tuning needs labelled data to tune against, and which labelled data is the most consequential decision in this section. Only three candidates exist. The project's own human-validated sample is one, and using it would be circular: the thresholds would be fitted to the same posts later used to judge the framework, so the agreement figure reported for RQ1 would be partly a measurement of that fitting. A pilot run over the target corpus is the second, but the target corpus carries no labels at all, so there is nothing there to tune against. The held-out expert set is the only candidate that is both labelled and independent of the evaluation. Tuning here is what keeps the gold set clean, and that is this stage's most important contribution to the design.

It checks Voter 1 for inherited bias. Section 3.3.2 identified the risk that Voter 1 carries a neutral skew from its training data, and Section 3.4.1 explained how a confident neutral skew would defeat the gate without ever announcing itself. Comparing Voter 1's predicted class distribution against expert labels on this set is what makes such a skew visible. Because the

check runs before the full annotation, the available response is to retrain or recalibrate, rather than to discover the problem inside a finished corpus.

It is a trial run. The full annotation run consumes a large amount of computing time. Running the complete pipeline end to end over the 600 held-out expert posts allocated below first shows whether the stages work together, what share of posts the gate routes to adjudication, and whether accuracy against expert labels is good enough to justify going further. If the design is poor, this is where it becomes apparent, at a cost measured in hours rather than days.

How the expert code-mixed posts are divided. The allocation is fixed here, not deferred, because no portion may serve two purposes and the pool is small. The corpus holds 2,320 code-mixed rows, of which 39 reached no majority among the three expert annotators and carry no sentiment label, leaving 2,281 usable. Of those, 800 fine-tune Voter 1, 600 are held out for threshold tuning and the trial run above, 120 form the annotator calibration gate in Section 3.6.2, and the remaining 761 form the secondary test set used in Section 3.7.2. The four portions are drawn in a single pass under a fixed random seed, stratified by sentiment class so that each carries the same class proportions as the pool, and are recorded before any stage runs so that no row can later be reassigned.

What this stage cannot do. Two limitations follow.

First, **the domain does not match**. The expert-annotated corpus is drawn from the public-security domain, while the target corpus is general social media, and the sentiment classes are distributed differently in the two. Thresholds tuned on one are being applied to the other. There is no clean fix, because no expert-annotated general-domain Manglish corpus exists at a usable size. The response is to state the limitation and to treat the tuned thresholds as starting values whose effect is tested by the sensitivity check described in Section 3.4.3, rather than as settled constants.

Second, the annotation is published but its reliability is not quantified. A peer-reviewed data paper accompanies the corpus (Md Suhaimin et al., 2024b) and records that three expert annotators from different fields labelled every post for sentiment and sarcasm, that the panel was sized at three so that a majority vote could settle a tie, and that 162 posts on which no majority formed carry no label. What that paper does not report is an inter-annotator agreement statistic, or the guideline the annotators worked from beyond a briefing on the task. How consistently the three worked therefore cannot be established from the published account. They remain the most reliable independent labels available for this text type, which is why the project uses them, but the case for them is weaker than a published agreement figure would provide, and the project does not overstate it.

3.6.2 Stage 2: Human Validation of Framework Output

Stage 2 compares the framework's output with human judgement, and produces the evidence for H1: a tri-model, **architecturally heterogeneous** majority-consensus framework with human-in-the-loop validation will produce sentiment labels that reach a substantial level of agreement with expert human annotation, higher than the agreement produced by single-model or same-architecture ensemble baselines.

The annotator gate comes first. Two bilingual annotators do the labelling, and their competence and independence are established before any labelling begins rather than assumed. Three controls do this.

Selection criteria stated in advance. Both annotators are native bilingual Malaysians who use Manglish for daily communication. Formal language qualifications such as SPM and MUET count as supporting evidence only, not as the criterion, because command of formal Malay and English is not the same thing as command of the informal mixed variety this corpus contains.

A calibration gate that must be passed before any real annotation begins. Each annotator labels a small set of posts drawn from the expert-annotated corpus, and their labels are compared against the three-expert labels. Both must reach $\kappa \geq 0.61$ against the three-expert labels, on Cohen's Kappa, raw percentage agreement and Gwet's AC1 alike, before they may proceed. That is the floor of the "substantial" band on the Landis and Koch scale, and requiring it on all three coefficients rather than on Kappa alone means an annotator cannot pass on a favourable class distribution. The threshold is set beforehand rather than chosen after the results are seen, and it is set with reference to what comparable work achieves. Kasmuri and Basiron (2019) report a Kappa of 0.83 on a Malay-English code-switching corpus. This turns annotator competence from something asserted into something measured.

Blinding. Neither annotator sees the machine label for any post. They label from the text alone and independently of each other, and the framework's labels are joined to theirs only after both have finished.

Taken together, annotator competence is measured against expert labels before annotation starts, and neither annotator can be influenced by the output being judged.

One limitation of the gate should be stated. Its calibration items come from the public-security domain, while the posts to be annotated come from general social media. The gate therefore tests command of the labelling scheme rather than familiarity with the target domain. That is what it is for, and it is worth saying so rather than implying more.

The annotation guideline. Both annotators work from a written guideline studied before labelling begins. Its core is the simple sentiment questionnaire proposed by Mohammad (2016):

three reminders (assess the language the speaker is using rather than the annotator's own view; label as most people would; identify the dominant sentiment inferable from the text), together with five response options covering positive language, negative language, sarcasm or ridicule, mixed language, and neither. To these it adds guidance specific to Manglish: discourse particles carry no polarity of their own; the negator appears as *tak*, *tk* or *x*; Malaysian senses of words such as *terror*, *power* and *best* differ from their standard English senses; and orthographic emphasis marks intensity rather than direction.

How sarcasm is recorded. Mohammad (2016) makes sarcasm, ridicule and mockery a response option in its own right, so that an annotator does not have to decide whether such a post is positive or negative, and leaves it to the application to settle how that option is mapped onto a label. The decision flowchart supplied with the guideline resolves it by inversion: a post using positive language that is sarcastic becomes negative, and the reverse. Sentiment and sarcasm form a single axis, and the reversal is mechanical. This project takes the other route the questionnaire allows and treats them as two axes: each post receives an independent sentiment label and a separate sarcasm flag, so sarcasm affects the sentiment through meaning rather than through an automatic flip.

The departure from the supplied flowchart is declared rather than adopted quietly, and three reasons support it. The corpus used for calibration records sentiment and sarcasm as two separate expert-annotated label sets, so annotators trained on a mechanical flip would consistently disagree with the reference on affectionate mockery, which is common in Manglish. The error analysis in Section 3.7 needs the sarcasm axis intact, and collapsing it into polarity destroys the variable that analysis depends on. And on clear cases the two schemes give the same answer; they diverge mainly on posts that are sarcastic in form but warm in intent, where the separate-axis reading is the more accurate one.

One property of the supplied flowchart is worth conceding, rather than defending the departure on every count. It never returns neutral for a post that uses positive language in part and negative language in part, resolving it instead to whichever dominates, which guards against neutral becoming the catch-all bucket that Chapter 2 criticises in existing corpora. The compensating control here is a "difficult" flag, which keeps such posts identifiable rather than quietly absorbed into neutral, and the number of neutral labels carrying that flag is reported alongside the results.

Which posts are annotated. The sample is stratified rather than random, along four dimensions, each with its own reason. The first is the predicted label, with the minority classes oversampled, because a random sample would follow the corpus distribution and might contain too few negative posts to support any reliable statement about that class. The second is the consensus route, meaning whether the post was accepted or adjudicated, with the branch that

sent it there recorded so that agreement can be reported per branch. Stratifying on it is what allows agreement to be reported separately for each route, so that the framework's parts can be assessed rather than only its output as a whole. A framework that agrees well overall but poorly on adjudicated cases is a different result from one that agrees evenly across routes, and only this stratification can tell the two apart. The third is the code-mixing band, using the measure computed in Section 3.2.3, so that agreement on lightly and heavily mixed text can be reported separately. The fourth is a per-author cap. Section 3.2.2 established that the corpus comes from 50 authors, and without a cap a sample of a thousand posts could be dominated by two or three prolific accounts, so that the result would describe those accounts rather than the corpus.

How large the sample is, and why. The sample holds at least 500 posts, and up to 1,000 where annotator time allows. The lower bound is the figure the project commits to, and it is derived rather than borrowed from convention.

H1 claims that the framework reaches a *substantial* level of agreement with human annotation. "Substantial" is a band on the Landis and Koch scale spanning 0.61 to 0.80, and each band on that scale is 0.20 wide. For the claim to be settled either way, the confidence interval around the measured Kappa must sit inside a single band rather than straddle a boundary. Otherwise, the result is equally consistent with two different conclusions. Table 3.9 gives the approximate half-width of a 95% confidence interval on Cohen's Kappa at a range of sample sizes.

Table 3.9: Approximate 95% confidence interval half-width for Cohen's Kappa

Posts annotated	**at κ = 0.61**	**at κ = 0.70**	**at κ = 0.80**
200	± 0.09	± 0.08	± 0.07
300	± 0.07	± 0.07	± 0.06
500	± 0.06	± 0.05	± 0.04
750	± 0.05	± 0.04	± 0.04
1,000	± 0.04	± 0.04	± 0.03

Half-widths are computed from the large-sample standard error of Cohen's Kappa given in Equation 3.1:

$$SE(\kappa) = \frac{1}{1 - p_e}\sqrt{\frac{p_o(1 - p_o)}{n}}$$

Equation 3.1: Large-sample standard error of Cohen's Kappa

Where p_o is the observed agreement between the two sets of labels, p_e is the agreement expected by chance, and n is the number of posts annotated. Each half-width in Table 3.9 is 1.96 standard errors. Three assumptions stand behind the table. The agreement expected by chance is taken as one third, because the sample is stratified toward balance across the three classes, and two annotators drawing at random from three equally likely categories agree one time in three; a sample further from balance would raise the expected chance agreement and widen the interval. The interval is the large-sample normal approximation at 1.96 standard errors, which is appropriate at the sample sizes in the table and would not be at much smaller ones. And the observed agreement corresponding to each Kappa is recovered from the relation the coefficient defines, so the table reports the interval a given Kappa would carry rather than one measured from data the project does not yet have. Half-widths are rounded to two decimal places.

At 200 posts the interval spans roughly 0.18, which is almost the full width of a band: a measured Kappa of 0.65 would carry an interval of about 0.56 to 0.74, crossing the boundary between moderate and substantial agreement, and H1 could not be settled in either direction. At 500 posts the interval narrows to about 0.12 and fits inside a band. At 1,000 it is around 0.08, which leaves a margin.

The upper end of the range is preferred for a second reason. Because the sample is stratified by consensus route, agreement is also reported for each route separately, and each of those figures is computed on a fraction of the sample. A sample large enough to support one overall figure is not automatically large enough to support three.

Which statistics are reported, and why more than one. Cohen's Kappa is the primary measure, because it expresses agreement after subtracting the agreement that would be expected by chance. Chapter 2 established that Kappa carries a specific weakness under imbalanced classes: a dominant class inflates the expected chance agreement, so high raw agreement can collapse into a misleadingly low Kappa (Feinstein & Cicchetti, 1990). This corpus is expected to be imbalanced, so Kappa reported alone could understate genuine agreement. It is therefore reported together with raw percentage agreement and the class prevalence, or accompanied by a prevalence-robust coefficient such as Gwet's AC1 (Gwet, 2008), so that a reader can tell whether a low figure reflects poor agreement or simply the distribution of the classes.

Two different agreements are measured, and they should not be confused. The first is between the two annotators, computed on their independent labels before any discussion takes place; this establishes that the gold set is itself trustworthy. The second is between the agreed human labels and the framework's labels, and that is the H1 result. Where the two annotators disagree, the disagreement is resolved by discussion to consensus, and that consensus label

enters the gold set, but it plays no part in the first figure, which is computed on the independent labels precisely so that discussion cannot inflate it.

The baselines H1 is measured against. H1 claims not only that the framework reaches substantial agreement with human annotation, but that it reaches closer agreement than the alternatives a reader would otherwise reach for. Both halves are therefore measured, on the same posts and with the same statistics. Three comparison figures are computed alongside the framework's own. The first is the single-model baseline: each voter's own label, taken before the gate, scored against the human labels, which is the agreement obtainable from one model used alone. The second is the same-architecture ensemble baseline: the majority label of Voters 2 and 3, the two decoders, with the encoder excluded, which is the agreement obtainable from a panel whose members share an architecture, the configuration Section 3.3.1 argues against. The third is the framework itself. Because Section 3.4.3 stores every voter's label and distribution on every post, none of the baselines requires further inference; each is recomputed from those records against the same gold set. Reporting the three together is what allows the second half of H1 to be settled rather than assumed, and a baseline that matched the framework would be a result the project would have to report.

3.6.3 The Dual Role of the Gold Evaluation Set

The same 500 to 1,000 human-labelled posts are used twice, for two different comparisons. Stating this plainly and early prevents a misreading that would otherwise be very damaging.

Table 3.10: The two comparisons the gold evaluation set serves

	What is compared	**Metric**	**Question answered**
RQ1	Human labels against the **framework's** labels	Cohen's Kappa, with raw agreement and class prevalence, or Gwet's AC1	Is the annotation framework producing labels a human would accept?
RQ2	Human labels against the **models'** predictions	Macro-F1	Which model classifies Manglish sentiment better?

Table 3.10 sets out the two comparisons. Same posts, same human labels, two different things compared against them. In RQ1 the human labels are the **standard** the framework is measured against. In RQ2 they are the **answer key** the models are marked against.

Saying this explicitly matters because the alternative reading is both damaging and easy to fall into. A reader who does not notice the reuse may assume the RQ2 models were scored against the framework's machine labels. That would be circular. It is not what happens: the models are marked against human labels throughout.

Why the set is reused rather than annotated twice. These are the most reliable labels the project holds. A second annotation round would cost the same effort and produce labels of the same quality on different posts, adding nothing to the reliability of either measurement.

One consequence follows. The gold set is stratified, not random. It deliberately over-represents minority classes, adjudicated cases and heavily mixed text. Two things follow.

Absolute Macro-F1 measured on this set **understates real-world performance** on the full corpus, because the set is harder than the corpus it is drawn from. A figure obtained here must never be reported as "model X achieves Y% on Manglish" without that qualification. Where an absolute figure is wanted, a reweighted estimate correcting the stratification back toward corpus proportions can be reported alongside the raw one.

Model ranking should nevertheless hold, and ranking is what H2 actually claims. H2 is a comparison between models rather than a statement about absolute accuracy, and every model faces the same test set, so the stratification makes the test harder for all of them equally.

A further consequence is that the minority classes will be thin in absolute terms. At 500 to 1,000 posts across three classes, the negative stratum may hold only a few hundred items, and Macro-F1 computed over that many is noisy.

3.6.4 Permitted Use of Each Dataset

The most likely objection to a project of this shape is that it marks its own work. The design's answer is that nothing used to build or tune the framework is later used to judge it. That answer is distributed across five sections, so Table 3.11 gathers it in one place where it can be checked at a glance.

Table 3.11: Permitted use of each dataset

Dataset	Voter 1 fine-tuning	Threshold tuning	Annotator calibration	RQ2 training	Evaluation
MESocSentiment	Candidate source, decided by comparison	No	No	No	No

Dataset	Voter 1 fine-tuning	Threshold tuning	Annotator calibration	RQ2 training	Evaluation
Expert corpus: fine-tuning portion	Yes	No	No	Yes, for adaptation	No
Expert corpus: Stage 1 held-out portion	No	**Yes**	No	No	No
Expert corpus: calibration portion	No	No	**Yes**	No	No
Expert corpus: secondary test portion	No	No	No	No	Yes, RQ2 secondary
Target corpus, unlabelled	No	No	No	No	No
Tier 2: machine-annotated corpus	No	No	No	**Yes**	No
Tier 3: gold evaluation set	No	No	No	No	**Yes, RQ1 and RQ2 primary**

Four readings of the table carry the argument.

The gold set's row is empty until the final column. It touches nothing that shapes the framework: not voter training, not threshold tuning, not annotator calibration, not the training data for the RQ2 models. It is used once, at the end, to measure.

Threshold tuning happens in exactly one place, and that portion of the expert corpus appears nowhere else in the table. This is the separation Section 3.6.1 argued for.

No portion of the expert corpus is used both to build the framework and to judge it. The fine-tuning portion does appear twice in the table, under Voter 1 fine-tuning and under RQ2 training, but both of those are building roles; no portion that shaped the framework reappears in the evaluation column. That is why the allocation in Section 3.6.1 is settled before any stage runs rather than decided as the project goes: the portions are drawn from one pool of 2,320 posts, and a portion crossing that line would put a hole in this table.

The target corpus contributes no labels to anything, because it has none. That is precisely why it was selected, as Section 3.2.1 explained, and it is the reason the framework exists at all.

3.7 Comparative Evaluation (RQ2)

Sections 3.2 to 3.6 describe how the corpus is built and how its quality is established. This section describes what the corpus is then used for. RQ2 asks how multilingual transformer models compare with monolingual baselines on highly code-mixed Malaysian text, and how robust each is to code-switching and noisy spelling. Answering it requires a comparison that is fair by construction, so that whichever way the result falls it can be trusted.

3.7.1 Models Compared and the Controlled Variable

Three models are compared: **XLM-RoBERTa, BERT-base** (Devlin et al., 2019) and **IndoBERT** (Wilie et al., 2020).

Why these three form a meaningful contrast. The text under study mixes English and Malay within single posts. BERT-base is a monolingual English model, so it covers one of the two languages. IndoBERT is monolingual in Indonesian, which is closely related to Malay, so it covers the other side of the mixture as closely as an available monolingual model can. XLM-RoBERTa is multilingual and covers both. The comparison therefore asks a specific question: on text that draws on two languages at once, does a model trained across many languages outperform models that each command only one side of the mixture?

That framing also sets the expectation honestly. Neither monolingual baseline is being set up to fail on unfamiliar material; each is strong on roughly half of every post. If XLM-R wins, it wins because handling both languages together matters more than handling one of them well.

The controlled variable. All three models are fine-tuned on identical training data, evaluated on identical test sets, and trained under the same procedure and the same set of random seeds. The model is the only thing that varies. Any difference in the result is therefore attributable to the model rather than to what it was shown.

What the models train on. The training data is the machine-annotated corpus the framework produces, the Tier 2 output described in Section 3.1.2, together with the Malay-only and English-only portions of the expert-annotated corpus and part of its code-mixed portion, used for linguistic adaptation. Training the RQ2 models on the corpus built in RQ1 is the point rather than a convenience: RQ1 produces the resource, and RQ2 demonstrates that the resource is useful for the task it was built for.

The same architecture is fine-tuned twice, and the two runs are not the same model. XLM-RoBERTa appears twice in this project: as the fine-tuned voter in Section 3.3.2, and as one of

the three models compared here. The architecture is the same in both places, but they are two separate fine-tuning runs producing two separate checkpoints, and they cannot be otherwise. Voter 1 is fine-tuned on the external corpora and must exist before the annotation run, because it helps produce the Tier 2 corpus. The model compared here is fine-tuned afterwards on that corpus, on exactly the same data as BERT-base and IndoBERT, because RQ2 requires the model to be the only thing that varies. Reusing Voter 1's checkpoint would hand XLM-R training data the two baselines never saw, and the comparison would no longer answer RQ2. The two runs differ in training data, in purpose, and in when they come into existence.

A residual advantage that must be stated. Even with separate checkpoints, XLM-R contributed one vote to every label in the corpus that all three RQ2 models train on. If XLM-R has systematic tendencies, some part of them is present in those labels, and XLM-R could then score higher than the baselines partly because the training labels resemble what an XLM-R produces, rather than because it handles code-mixed text better. That is a genuine threat to H2 and it is not fully removable. Three things limit it, and one measures it.

The models are marked against **human** labels, not machine labels, so the advantage can only operate through the training data and never through the answer key. The gate in Section 3.4 means an accepted label required XLM-R to agree, confidently, with at least one confident voter from a different lineage, so XLM-R never determines a label on its own. And the second test set described in Section 3.7.2 consists of expert-labelled posts that this framework never annotated at all, so if XLM-R's margin is large on the primary test set and absent on the second, that pattern is evidence the margin came from label provenance rather than from competence. The two-test-set design is therefore partly a diagnostic for this confound, not only a generalisation check, though the domain difference discussed below means it is an indication rather than a clean measurement.

3.7.2 Split Design and the Author-concentration Constraint

The separation rule, in operational terms. No post used to train or tune any part of the project appears in any test set, and no author appears on both sides of a split. Author-level separation is the stricter of the two conditions and the one that matters here, because Section 3.2.2 established that the target corpus averages about 14,180 posts per author. At that density a model has ample material to learn who wrote a post alongside what the post means, and a split made post by post would let an author's habits appear in both training and test data, inflating the result without any gain in sentiment competence.

The constraint this runs into. The target corpus has 50 authors. An author-disjoint split therefore divides 50 units, not thousands. A conventional split would put perhaps forty authors in training and ten in test, which rests a claim about Malaysian code-mixed sentiment on ten

writers. Per-author style effects at that scale are large, and a difference between two models measured on ten people is not securely a difference about the language.

The design chosen in response. Rather than a single split, the evaluation cross-validates across author folds: the 50 authors are divided into five folds of ten, and each fold takes a turn as the test set while the remaining forty train. Every author is therefore tested on exactly once, the result rests on all 50 rather than on ten, and the variation between folds becomes visible instead of hidden. Per-author variance is reported alongside the mean, because a model whose performance swings widely between writers is a different result from one that performs evenly, and the swing itself is informative about how much the figure depends on who is writing.

A second test set with broad author coverage. The code-mixed portion of the expert-annotated corpus carries 1,979 distinct authors across 2,320 posts, almost the opposite profile to the target corpus, which has depth without breadth. A held-out portion of it, never used in training, therefore serves as a secondary test set. A model that performs comparably on both test sets is generalising past the writing habits of individual authors. A large gap between them is itself an informative result. Table 3.12 sets out the full arrangement.

Table 3.12: Data flow for the RQ2 evaluation

Role	Source	Split rule
Training	The machine-annotated corpus (Tier 2), plus the Malay-only and English-only portions of the expert corpus and part of its code-mixed portion, for linguistic adaptation	Author-disjoint from both test sets
Test set 1, primary	The human-validated gold set (Tier 3), drawn from the annotated target corpus the same set used for the RQ1 agreement measurement, see Section 3.6.3	Author-disjoint from training, with a per-author cap
Test set 2, generalisation	A held-out portion of the expert corpus's code-mixed slice, never used in training	Author-disjoint from the expert-corpus rows used in training

Why the expert corpus appears in both training and testing, which looks wrong but is not. The split is by author, and that corpus has 1,979 authors in its code-mixed portion, so an author-disjoint split within it is straightforward. Keeping it in training is in fact the better choice for this purpose, because it holds the domain constant across the two roles. A drop in performance

on the held-out portion is then attributable to unseen authors rather than to unseen domain. Excluding it from training entirely would make any drop ambiguous between the two explanations.

The confound that remains. The two test sets differ on two axes at once, not one. The primary set comes from 50 general-domain authors; the secondary set comes from roughly 2,000 public-security-domain authors. A gap between the two therefore cannot be attributed to author diversity alone, because domain shift is a competing explanation and the design cannot separate them. Comparable performance across both remains meaningful evidence of generalisation. A large gap is informative but ambiguous, and it will be reported that way rather than as a clean author effect.

What can be claimed, and what cannot. The RQ2 result generalises across these authors and this text type. It does not establish that the same ordering of models would hold across Malaysian social media as a whole, because no corpus exists that would allow that to be tested. Broader generalisation is future work, and the chapter says so rather than letting the scale of the corpus imply a breadth the sample does not support.

3.7.3 Metrics and Reporting Protocol

Macro-F1 is the primary metric, and accuracy is not used. Accuracy counts every post equally, so on an imbalanced corpus it rewards a model for predicting the majority class and ignoring the rest. That is exactly the failure Chapter 2 identified in existing corpora, and a metric that fails to penalise it would be the wrong instrument for a project built to correct it. Macro-F1 averages the F1 score of each class with equal weight, so a model that performs well on neutral and poorly on negative cannot hide the second behind the first. This holds even though the primary test set is stratified toward balance, because the secondary test set is not, and because the project's concern is minority-class performance regardless of how the test set happens to be composed.

Robustness slices answer the second half of RQ2. RQ2 asks not only how the models compare but how robust they are to code-switching and noisy spelling. Results are therefore reported separately by code-mixing band, using the measure computed in Section 3.2.3, and by a noisy-spelling indicator that records how far a post's spellings depart from standard forms.

These slices are not supplementary reporting. H2 makes a specific prediction: that the difference between multilingual and monolingual models is **greatest** on posts marked by heavy code-switching and noisy spelling, and only a sliced result can test that prediction. An overall Macro-F1 figure, however carefully produced, would leave the central claim of H2 untested.

Several random seeds, with the spread reported. Each model is trained under at least three random seeds, and five where time allows, and the results are reported as a mean with a standard deviation, never as a single figure. Fine-tuning is stochastic, and two runs of the same model on the same data produce different results. A one-point gap between two models from a single run each is not evidence of a difference between the models; it may be smaller than the variation between runs of either one. Reporting the spread is what allows a reader to judge whether an observed gap is larger than the noise.

There is a second and independent reason. Section 3.6.3 noted that the minority classes in the gold set will be thin in absolute terms, so Macro-F1 computed over them is noisy for reasons that have nothing to do with the models. Both sources of variation argue for the same protocol.

H2 is a neutral comparison. The project does not need XLM-R to win. Chapter 2 established that the superiority of multilingual models on code-mixed text is widely assumed but not demonstrated for this language pair, and gave reasons to doubt it: the curse of multilinguality, and the under-representation of Malay in multilingual pre-training. A result showing that a monolingual baseline matches or beats XLM-R would answer RQ2 just as well as one showing the opposite, and would be the more interesting finding of the two. What would make the evaluation fail is not a particular direction of result but an unsound comparison, which is why the controls in Sections 3.7.1 and 3.7.2 carry more weight in this design than the outcome does.

Error analysis on consensus failures. The expert-annotated corpus carries sarcasm labels alongside its sentiment labels. Section 3.3.1 identified sarcasm as the most likely place for the two generative voters to share a blind spot, and these labels allow that to be examined directly: whether agreement among the voters, and accuracy of the resulting labels, degrade on sarcastic posts relative to the rest. This is reported as error analysis rather than as a hypothesis test, since the project makes no prediction about it, but it is the one place where the residual heterogeneity risk conceded in Section 3.3.1 can be measured rather than argued about.

3.8 Experimental Environment and Reproducibility

This section sets out the environment the work runs in, and what the project pins and publishes, so that another researcher can reproduce the work rather than take its results on trust. A design that cannot be repeated is a description rather than a method.

3.8.1 Hardware and Software Environment

Every stage of this project runs on a single workstation, and its specification is stated here because two of the design decisions in this chapter follow from it rather than from preference.

The machine carries an Intel i9-11900 with sixteen logical cores, 127.9 GB of system memory, an NVIDIA RTX 2080 SUPER with 8 GB of video memory, and 178 GB of free storage. It is shared with other students and is unavailable for fourteen hours in each week, leaving approximately 154 hours available.

The binding constraint is the 8 GB of video memory, and nothing else. After display and remote-access overhead, roughly 7.0 to 7.5 GB is usable. System memory, processor and storage are all comfortable at the scale this project works at; the graphics memory is not. Two decisions elsewhere in this chapter follow directly from it. Section 3.3.3 runs the generative voters one after another rather than together, because the card holds one at a time. Section 3.5.1 selects an adjudicator that fits entirely within the card at four-bit quantisation, and states openly that this is why the adjudicator does not outrank the voters in scale.

The card's architecture rules out three things that a reader may otherwise assume were used. It does not support bfloat16, so training runs in float16 with automatic mixed precision. It does not support FlashAttention-2, so attention uses the memory-efficient backend available in the framework instead. And it does not support the newer quantisation kernels, so four-bit quantisation uses the methods that the architecture does support. Naming these matters for reproduction: a reader running the same configuration on newer hardware would obtain different memory behaviour and different timings, and would otherwise have no way of knowing why.

Table 3.13 sets out the compute budget the project is planned against. The figures are estimates until the pilot in Section 3.1.2 replaces them with measurements, and they are given as ranges for that reason.

Table 3.13: The compute budget the project is planned against

Stage	Job	Estimated GPU hours
Voter 1 comparison	Three training conditions, three seeds each	8 to 15
Pilot	10,000 posts through all three voters	2 to 4
Full annotation	The filtered corpus through all three voters	35 to 70
Adjudication	The routed posts only	4 to 8
RQ2 evaluation	Three models, three to five seeds each	10 to 20
Reserve	One complete re-run of the pipeline	About 50
Total		109 to 167

The total is roughly three-quarters of a week to just over one week of compute. The margin is why the design commits to a floor rather than to a ceiling, and why a shortfall would reduce the size of the result rather than force a change to the design.

3.8.2 Running the Annotation

The annotation run takes days and shares a machine with other users, so three operational decisions are part of the method, and are stated here for that reason.

The run is resumable by design. Output is written to append-only shards keyed by the stable row identifier assigned in Section 3.2.3, and on starting, the run skips identifiers already written. Nothing is held in memory that would be lost on an interruption, and an interruption therefore costs one shard of work.

The run yields the machine when it is booked by someone else. The unavailable windows are held as configuration, not written into the code, and when the next window is within a few minutes the run flushes and exits instead of pausing, because a paused process keeps the graphics memory reserved and the machine unusable to the person who booked it. It restarts afterwards from the shards already written.

The configuration is verified before the long run. A short run is completed end to end first, because a job on this class of hardware can be terminated by an operating-system event that has nothing to do with the job itself, and discovering that after forty hours costs forty hours.

Every model is used at a pinned release, not referred to by name. Table 3.14 records what is pinned, and Section 3.8.3 explains why naming a model is not sufficient.

Table 3.14: The pinned release for every model in the pipeline

Role	Model	Precision	Specified in
Voter 1	XLM-RoBERTa base	Full precision, fine-tuned	§3.3.2
Voter 2	Llama 3.1 8B	Four-bit	§3.3.3
Voter 3	Qwen2.5 7B Instruct	Four-bit	§3.3.3
Adjudicator	SEA-LION, Gemma 2 9B lineage	Four-bit	§3.5.1
RQ2 model	XLM-RoBERTa base (separate checkpoint)	Full precision, fine-tuned	§3.7.1

Role	Model	Precision	Specified in
RQ2 baseline	BERT-base	Full precision	§3.7.1
RQ2 baseline	IndoBERT	Full precision	§3.7.1

3.8.3 What is Pinned and What is Published

What is pinned. Every model in the pipeline is used at a specific released version, recorded rather than referred to by name alone. Model names move: a checkpoint updated six months after publication is a different model carrying the same name, and a result obtained against the earlier one cannot be reproduced against the later one. The quantisation method is recorded for the same reason, since a model run at reduced precision does not behave identically to the full-precision version, and naming the model alone does not say which was used.

What is published. The verbatim prompts given to the generative voters and to the adjudicator; the per-voter confidence thresholds together with the procedure that produced them; the random seeds; and the configuration files for every training and inference run. Two of these deserve comment.

The prompts, because they are part of the method rather than an implementation detail. A generative model's answer depends on how it was asked, and small differences in wording produce different behaviour. A prompt that is described rather than quoted is a step a reader cannot repeat, so publishing them verbatim is what separates a described pipeline from a reproducible one.

The thresholds, together with the procedure that set them. A threshold value on its own is a number a reader must accept. Published alongside the tuning procedure and the data it was tuned against, it becomes a decision another researcher can inspect, disagree with, or redo. The sensitivity check described in Section 3.4.3 is reported for the same purpose.

The cached probability distributions. Section 3.4.3 records the full set of probabilities from every voter on every post rather than only the labels chosen. Releasing this alongside the labels means the routing rule can be re-run by anyone using different thresholds, without repeating the expensive inference stage. A reader who disagrees with the gate's settings can test their own for the cost of re-adjudicating the cases that change path, instead of running three models over the whole corpus again.

The label release. The project releases labels keyed to row identifiers rather than republishing the posts themselves. This is what the stable identifier assigned in Section 3.2.3 exists for.

Two reasons support this form of release. The first is that the terms under which the source corpus may be redistributed are not yet confirmed, as Section 3.2.1 noted. A label-only release stays within whatever those terms turn out to permit: the labels are the project's own contribution, and anyone wishing to use them obtains the posts from the original source under the original conditions. The second is that the same arrangement respects the conditions attached to the expert-annotated corpus, which is released under a CC BY-NC licence: its source must be acknowledged in any published work drawing on it, and its use is restricted to non-commercial purposes, both of which this project observes.

The cost of a label-only release should be acknowledged rather than passed over: it is less convenient than a single downloadable file, and it requires a user to obtain the source separately and join the two. It works, because the identifier joins the labels back to the source without ambiguity, and it means the release does not depend on a permission the project does not yet hold.

The reframe. Section 3.5.1 ruled out commercial services for the adjudicator and described the decision as a strength the project claims rather than a restriction it accepts. This section is where that claim is settled rather than asserted. Because every model in the pipeline is an open-weight model pinned to a specific release, and because every prompt, threshold, seed and configuration file is published, the whole framework can be regenerated by a third party with no budget beyond the hardware to run it.

Had the adjudicator been a commercial service, none of this would hold. The most consequential stage of the pipeline would have depended on a model that could change without notice, that carries no version a reader can pin, and that someone without a subscription could not run at all. The reproducibility argument in this section would then have had a gap at exactly the point where the framework does its hardest work.

What began as a constraint imposed by cost is therefore the property that makes the work reproducible by someone else. That is not a rationalisation constructed afterwards: it is the same reasoning that produced the decision in Section 3.5.1, stated here in the place where its consequence becomes visible.

3.9 Chapter Summary and Evaluation

This chapter set out the design of the Heterogeneous Ensemble Annotation Framework and the evaluation built around it. This section states what each objective has been specified by, and which section did the work, before turning to what the design cannot do.

Objective 1: collecting and preprocessing the corpus. Section 3.2 specified this. The target corpus was selected for a property that would normally be a shortcoming, namely that it carries no sentiment labels at all, which is what makes it suitable for a project whose purpose is to produce them. Its composition was reported openly, including the concentration of 709,012 posts among 50 authors, and the consequences of that concentration were carried forward to the sections that deal with them rather than left as a caveat. Filtering was kept light and its scope explained: because the corpus is code-mixed throughout, its size is bounded by available computing time rather than by data quality, and Section 3.1.2 commits to a floor of 150,000 annotated posts so that the RQ1 claim does not depend on that budget. The objective refers to normalisation of noisy spelling where appropriate, and Section 3.2.4 determined that it is largely not appropriate here. The conclusion is argued from the corpus's own two shipped files rather than assumed, since the version to which the standard preprocessing recipe had been applied lost the emoji, capitalisation and punctuation that carry sentiment in this variety, collapsed tens of thousands of distinct posts into duplicates, and dropped 58,603 posts without record.

Objective 2: designing the annotation framework. Sections 3.3 to 3.5 specified this. Section 3.3 defined the panel's heterogeneity by properties that predict independent failure: architecture, pre-training lineage and tokenisation. It did so rather than by implementation differences, having shown that implementation differences alone were what failed in the framework Chapter 2 criticised. Section 3.4 set out confidence-gated routing, the refinement this project claims as its contribution, under which agreement is accepted only when it is also confident. Section 3.5 specified a different-lineage, regionally adapted reasoning adjudicator for the cases the panel does not settle, and established that no post is discarded.

Objective 3: measuring agreement with human annotation. Section 3.6 specified this. Validation runs in two stages with distinct purposes: one before the annotation run, on expert-labelled data the project did not produce, which is where the confidence thresholds are set and where Voter 1's inherited bias would surface; and one after it, on the framework's own output. The human validation is preceded by a calibration gate that turns annotator competence from an assertion into a measurement. Its sample is stratified along four dimensions, each with a stated reason. Its size is derived from the precision needed to place a Kappa estimate inside a Landis and Koch band, not adopted by convention. Its statistics are chosen for the imbalanced conditions Chapter 2 identified. Table 3.11 makes the separation between building and judging auditable in one place.

Objective 4: evaluating multilingual against monolingual models. Section 3.7 specified this. Three models are compared on identical training data with the model as the only variable, split by author rather than by post and cross-validated across author folds so the result rests on all

50 writers rather than ten, and tested on two sets with opposite author profiles. Macro-F1 is the primary metric, reported across several random seeds with the spread shown, and sliced by code-mixing band and noisy-spelling indicator, which is where H2's specific prediction is actually tested.

Table 3.15 gathers the quantities and the model choices this chapter commits to. Each entry is a floor, not a forecast: each is set below what the available hardware is expected to deliver, so that a run falling short would reduce the size of the result without requiring the design to change.

Table 3.15: The minimum the design commits to for each open decision

Decision	**Committed minimum**	**If the run goes well**
Posts annotated (Tier 2, the RQ1 deliverable)	At least 150,000	The full filtered corpus (§3.1.2)
Gold evaluation set (Tier 3)	At least 500 posts	Up to 1,000 where annotator time allows (§3.6.2)
Voter 1	XLM-RoBERTa, fine-tuned	Training mix settled by the comparison (§3.3.2)
Voters 2 and 3	Llama 3.1 8B and Qwen2.5 7B Instruct	A smaller release from the same developer if either will not run (§3.3.3)
Adjudicator	SEA-LION, Gemma 2 9B lineage, four-bit	A smaller SEA-LION release, never a voter's family (§3.5.1)
Expert code-mixed pool (2,281 usable of 2,320)	800 fine-tuning, 600 held-out, 120 calibration, 761 secondary test	Split drawn once and then fixed (§3.6.1)
Human annotators	Two bilingual annotators	Both must pass the calibration gate before labelling (§3.6.2)
Models compared for RQ2	XLM-RoBERTa, BERT-base and IndoBERT	Identical training data, the model the only variable (§3.7.1)
RQ2 cross-validation	Five author folds of ten	Every author tested on exactly once (§3.7.2)
RQ2 random seeds	At least three per model	Five where time allows (§3.7.3)

Decision	Committed minimum	If the run goes well
H1 baselines	Each voter alone, and the two-decoder majority, scored on the gold set	Recomputed from the stored distributions, no extra inference (§3.6.2)

Five weaknesses are carried by this design, and none is answered here with an assurance that it will not matter. Each has a stated mitigation and a measurement that will show whether the mitigation worked. Table 3.16 sets them out.

Table 3.16: Limitations of the design, with mitigations and the measurement that tests each

Limitation	Mitigation	How it will be measured
Two of the three voters are decoder-only models and may share a blind spot, most plausibly on sarcasm (§3.3.1)	The gate does not accept agreement alone; adjudication comes from a third lineage; Stage 1 checks the panel before the full run	Agreement and accuracy reported separately on sarcastic posts, using the expert corpus's sarcasm labels (§3.7.3)
The adjudicator does not outrank the voters in scale, so it may not settle what the panel could not (§3.5.2)	Lineage separation and regional adaptation replace capacity as the justification	Adjudication accuracy on the routed cases, reported by route as a distinct result (§3.6.1)
Confidence thresholds are tuned on public-security-domain data and applied to general social media (§3.6.1)	Thresholds treated as starting values rather than settled constants	Threshold sensitivity check, made inexpensive by the cached distributions (§3.4.3)
The target corpus draws on only 50 authors, so an author-disjoint split has few units (§3.2.2)	Author-fold cross-validation rather than a single split; per-author cap on the gold sample; a secondary test set with roughly 2,000 authors	Per-author variance reported with the mean; the gap between the two test sets (§3.7.2)
XLM-RoBERTa cast a vote on the labels all three RQ2 models train on, which may advantage it (§3.7.1)	Evaluation against human labels only; the gate prevents any voter from fixing a label alone	Comparison of XLM-R's margin across the two test sets, the second of which this framework never annotated (§3.7.1)

A sixth limitation cannot be measured directly. The data paper accompanying the expert-annotated corpus used for threshold tuning and annotator calibration (Md Suhaimin et al., 2024b) records that three expert annotators labelled every post and that a majority vote fixed

each label, but it reports no inter-annotator agreement statistic and no annotation guideline beyond a briefing, so how consistently those three annotators worked cannot be established from the published account. The labels remain the most reliable independent labels available for this text type, which is why the project relies on them. The annotator calibration gate in Section 3.6.2 gives an indirect reading: if two independent bilingual annotators reach substantial agreement with those labels, that is evidence the labels are sound, and if they do not, the gate will show it before any annotation begins.

What these limitations have in common is more useful than any of them individually. Each is met by a measurement rather than by an argument, and each measurement is scheduled at a point where a poor result can still change the design, that is, before the full annotation run, not after it. A design whose weaknesses are measured is in a stronger position than one whose weaknesses have not been examined, and stating them here is what allows Chapter 4 to report against them rather than around them.

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